

A Pannon Egyetem gyakorlatorientált infrastrukturális- és készségfejlesztési reformja

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NUMERICAL ANALYSIS

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Numerical Analysis

2 Nonlinear Algebraic Equations and Systems

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2.1 Fixed-Point Iteration

A recursion of the form

$$p_{k+1} = h(p_k, p_{k-1}, \dots, p_{k-m+1}), \quad k \geq m-1$$

is called ***m*-order recursion** or ***m*-step iteration**. An *m*-step iteration is well-defined, if *m* number of initial values

$$p_0, p_1, \dots, p_{m-1}$$

are given.

Given a function $g: I \rightarrow I$, where $I \subset \mathbb{R}$. The recursive sequence

$$p_{k+1} = g(p_k), \quad k \geq 0,$$

which corresponds to an initial value $p_0 \in I$ is called a **fixed-point iteration**.

Example

Consider the function $g(x) = -\frac{1}{8}x^3 + x + 1$. The corresponding fixed-point iteration is

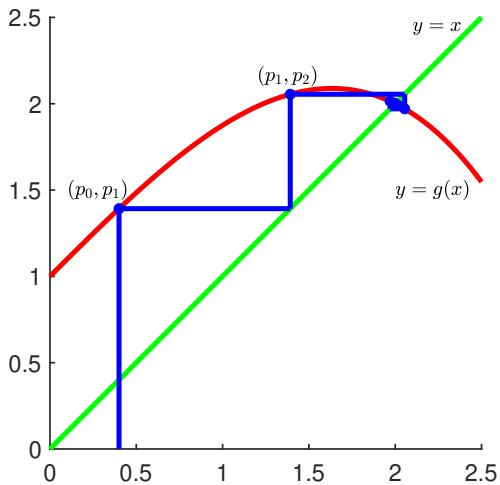
$$p_{k+1} = -\frac{1}{8}p_k^3 + p_k + 1.$$

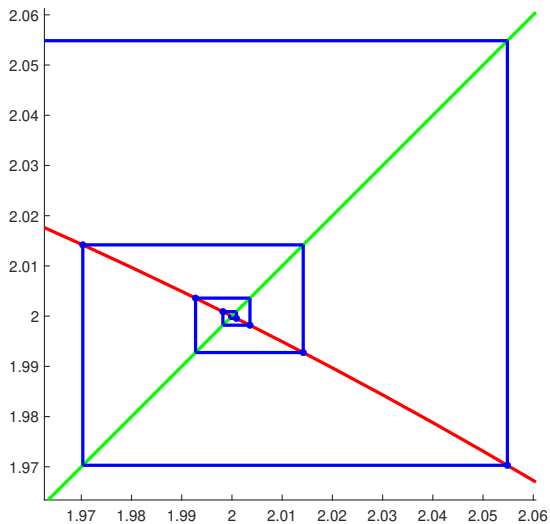
We compute several terms of the sequence starting from the initial condition $p_0 = 0.4$. From the next table it can be seen that the sequence p_k converges to a limit.

Fixed-point iteration $p_{k+1} = -\frac{1}{8}p_k^3 + p_k + 1$

k	p_k
0	0.40000000
1	1.39200000
2	2.05484646
3	1.97030004
4	2.01419169
5	1.99275275
6	2.00358428
7	1.99819822
8	2.00089846
9	1.99955017
10	2.00022477
11	1.99988758
12	2.00005620
13	1.99997190
14	2.00001405
15	1.99999297

stair step diagram or Cobweb diagram of the fixed-point iteration:





In the previous example we observed that the fixed-point iteration converged to the first coordinate of the intersection of the graphs of the line $y = x$ and the function $y = g(x)$. The first (and also the second) coordinate of this point satisfies the **fixed point equation**

$$g(x) = x.$$

The number p is called the **fixed point** of the function g , if it satisfies

$$g(p) = p.$$

Theorem:

Let $g: [a, b] \rightarrow [a, b]$ (or $\mathbb{R} \rightarrow \mathbb{R}$) be a continuous function, $p_0 \in [a, b]$ be fixed, and consider the fixed-point iteration

$$p_{k+1} = g(p_k).$$

If p_k is convergent and $p_k \rightarrow p$, then

$$p = g(p).$$

Proof

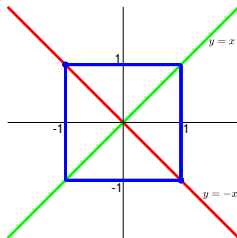
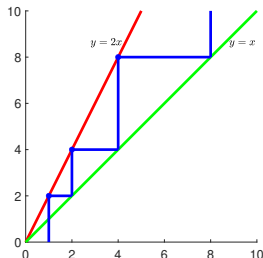
Using the continuity of g and the assumption $p_k \rightarrow p$ we get

$$\begin{array}{ccc} p_{k+1} & = & g(p_k) \\ \downarrow & & \downarrow \\ p & = & g(p). \end{array}$$

A fixed-point iteration is not always convergent, or the limit is not necessary finite.

Example

Consider the function $g(x) = 2x$. Then $p_{k+1} = g(p_k)$ and the initial value $p_0 = 1$ yield $p_k = 2^k$, which converges to infinity.



Example

Consider $g(x) = -x$. Then $p_{k+1} = g(p_k)$ and the initial value $p_0 = 1$ yield $p_k = (-1)^k$, which is not convergent.

We will use the short notations:

$$C[a, b] = C([a, b], \mathbb{R}) = \{f: [a, b] \rightarrow \mathbb{R}, f \text{ is continuous}\}$$

$$C^1[a, b] = C^1([a, b], \mathbb{R}) = \{f: [a, b] \rightarrow \mathbb{R}, f \text{ is continuous differentiable}\}$$

$$\begin{aligned} C^n[a, b] &= C^n([a, b], \mathbb{R}) \\ &= \{f: [a, b] \rightarrow \mathbb{R}, f \text{ is } n\text{-times continuously differentiable}\} \end{aligned}$$

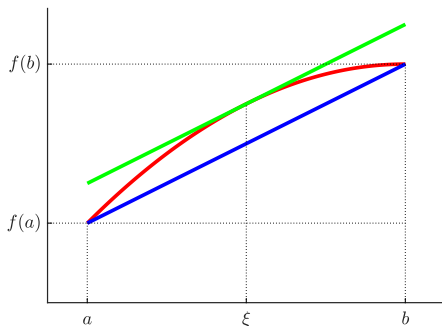
Theorem (Lagrange's Mean Value Theorem):

Let $f \in C^1[a, b]$. Then there exists $\xi \in (a, b)$ such that

$$f(b) - f(a) = f'(\xi)(b - a),$$

or equivalently,

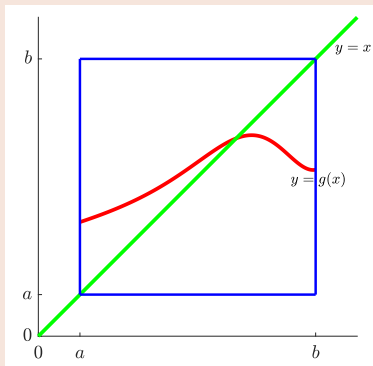
$$\frac{f(b) - f(a)}{b - a} = f'(\xi).$$



Theorem:

Let $g: [a, b] \rightarrow [a, b]$ be continuous. Then g has a fixed point in the interval $[a, b]$. Moreover, if g is differentiable on (a, b) , and there exists a constant $0 \leq c < 1$ such that $|g'(x)| \leq c$ for all $x \in (a, b)$, then this fixed point is unique.

Proof



Proof cont.

For the proof of the uniqueness, suppose that g has two fixed points p and q . Then it follows from the Lagrange's Mean Value Theorem that there exists a $\xi \in (a, b)$ such that

$$|p - q| = |g(p) - g(q)| = |g'(\xi)||p - q| \leq c|p - q|.$$

But this yields that $p = q$, i.e., the fixed point is unique.

Theorem (fixed-point theorem):

Let $g: [a, b] \rightarrow [a, b]$ be continuous, g is differentiable on (a, b) , and suppose that there exists a constant $0 \leq c < 1$ such that

$$|g'(x)| \leq c, \quad x \in (a, b).$$

Let $p_0 \in [a, b]$ arbitrary, and

$$p_{k+1} = g(p_k), \quad k \geq 0.$$

Then the sequence p_k converges to the unique fixed point p of the function g ,

$$|p_k - p| \leq c^k |p_0 - p|, \tag{1}$$

and

$$|p_k - p| \leq \frac{c^k}{1 - c} |p_1 - p_0|. \tag{2}$$

Proof

We know that g has a unique fixed point p . Since $0 \leq c < 1$ by our assumptions, the convergence $p_k \rightarrow p$ follows from (1). To show (1), we have from the assumptions and the Lagrange's Mean Value Theorem that

$$|p_k - p| = |g(p_{k-1}) - g(p)| = |g'(\xi)| |p_{k-1} - p| \leq c |p_{k-1} - p|.$$

Now mathematical induction gives relation (1) easily.

Proof cont.

To prove (2), let $m > k$ be arbitrary. Then the triangle inequality, the Mean Value Theorem and our assumptions imply

$$\begin{aligned}
 & |p_k - p_m| \\
 & \leq |p_k - p_{k+1}| + |p_{k+1} - p_{k+2}| + \cdots + |p_{m-1} - p_m| \\
 & \leq |g(p_{k-1}) - g(p_k)| + |g(p_k) - g(p_{k+1})| + \cdots + |g(p_{m-2}) - g(p_{m-1})| \\
 & \leq c|p_{k-1} - p_k| + c|p_k - p_{k+1}| + \cdots + c|p_{m-2} - p_{m-1}| \\
 & \leq (c^k + c^{k+1} + \cdots + c^{m-1})|p_0 - p_1| \\
 & = c^k(1 + c + \cdots + c^{m-k-1})|p_1 - p_0| \\
 & \leq c^k \sum_{i=0}^{\infty} c^i |p_1 - p_0|.
 \end{aligned}$$

Hence

$$|p_k - p_m| \leq \frac{c^k}{1 - c} |p_1 - p_0|, \quad m > k.$$

Keeping k fixed and tending with m to ∞ , we get (2).

We say that the function $g: I \rightarrow \mathbb{R}$ is **Lipschitz continuous** on the interval I , or other words, it has the **Lipschitz property**, if there exists a constant $c \geq 0$ such that

$$|g(x) - g(y)| \leq c|x - y|, \quad x, y \in I. \quad (3)$$

The constant c in (3) is called the **Lipschitz constant** of the function g .

- g is Lipschitz continuous on $I \implies g$ is continuous on I
- $g \in C^1[a, b] \implies g$ is Lipschitz continuous on $[a, b]$

$$|g(x) - g(y)| = |g'(\xi)||x - y| \leq c|x - y|,$$

where $c := \max\{|g'(x)| : x \in [a, b]\}$.

- g is piece-wise continuously differentiable $\implies g$ is Lipschitz cont.
One example is the function $g(x) = |x|$.

If g is Lipschitz continuous with a Lipschitz constant $0 \leq c < 1$, then g is called a **contraction**.

Theorem (contraction principle):

Let the function $g: [a, b] \rightarrow [a, b]$ be a contraction, $p_0 \in [a, b]$ be arbitrary, and $p_{k+1} = g(p_k)$ ($k \geq 0$). Then the sequence p_k converges to the unique fixed point p of the function g , and relations (1) and (2) are satisfied.

We say that the iteration

$$p_{k+1} = h(p_k, p_{k-1}, \dots, p_{k-m+1})$$

converges locally to p if there exists a constant $\delta > 0$, such that for every initial value

$$p_0, p_1, \dots, p_{m-1} \in (p - \delta, p + \delta)$$

the corresponding sequence p_k converges to p .

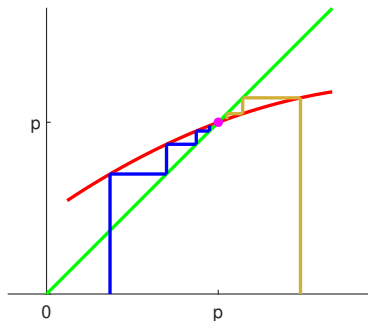
If the iteration p_k converges to p for every initial value, then this iteration method is called **globally convergent**.

Theorem:

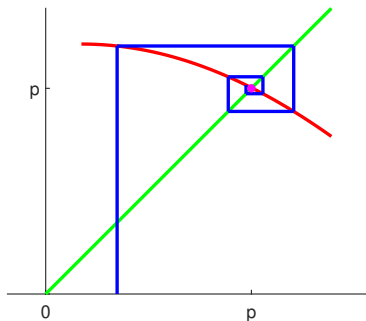
Let $g \in C^1[a, b]$, and let $p \in (a, b)$ be a fixed point of g . Suppose also that

$$|g'(p)| < 1.$$

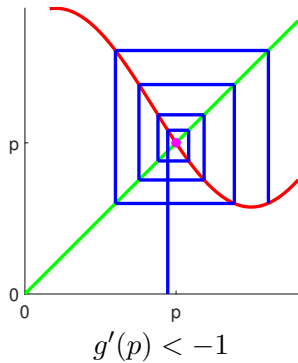
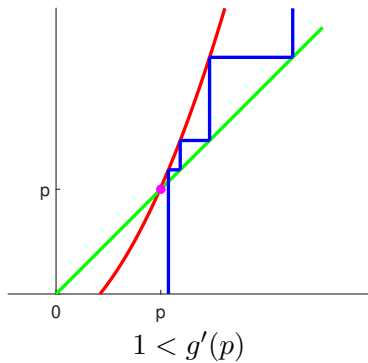
Then the fixed-point iteration **converges locally** to p , i.e., there exists a $\delta > 0$ such that $p_{k+1} = g(p_k)$ converges to p for all $p_0 \in (p - \delta, p + \delta)$.



$$0 < g'(p) < 1$$



$$-1 < g'(p) < 0$$



2.2 Stopping Criteria of Iterations

Suppose the sequence p_k tends to the limit p . We approximate the limit p by a term of the sequence p_k , where k is “large enough”. Question: how to determine the number of steps k for which p_k gives us a good approximation of p , where $f(p) = 0$.

We predefine three tolerances $\varepsilon_1 > 0$, $\varepsilon_2 > 0$ and $\varepsilon_3 > 0$. We consider the k th term p_k as an appropriate approximation of p if

$$(i) \quad |p_k - p_{k-1}| < \varepsilon_1, \quad (ii) \quad \frac{|p_k - p_{k-1}|}{|p_k|} < \varepsilon_2, \quad \text{or} \quad (iii) \quad |f(p_k)| < \varepsilon_3.$$

Condition (i) is a numerical analogue of the absolute error $|p_k - p|$ of the approximation.

Condition (ii) is the numerical analogue of the relative error $|p_k - p|/|p|$ of the approximation.

Condition (iii) tests whether the function value at p_k is close to 0.

Example

Consider the sequence

$$p_k = 1 + \frac{1}{2} + \cdots + \frac{1}{k}.$$

We have $|p_k - p_{k-1}| = \frac{1}{k}$, so condition (i) holds if k is large enough. On the other hand, $p_k \rightarrow \infty$ as $k \rightarrow \infty$.

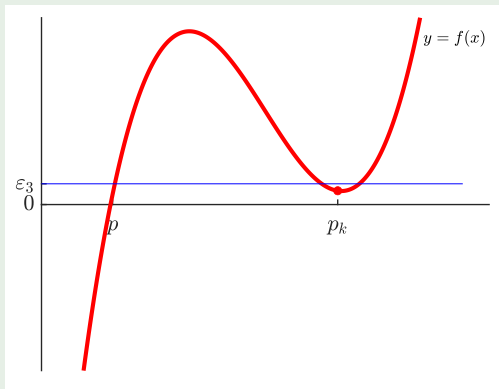
Similarly,

$$\frac{|p_k - p_{k-1}|}{|p_k|} = \frac{\frac{1}{k}}{1 + \frac{1}{2} + \cdots + \frac{1}{k}} \leq \frac{1}{k} \rightarrow 0, \quad \text{as } k \rightarrow \infty,$$

so condition (ii) holds for large k , but the sequence is not convergent.

Example

Consider the function with graph



Here condition (iii) holds, but p_k is not close to the root.

In practice we use combination of stopping criteria.

2.3 Bisection Method

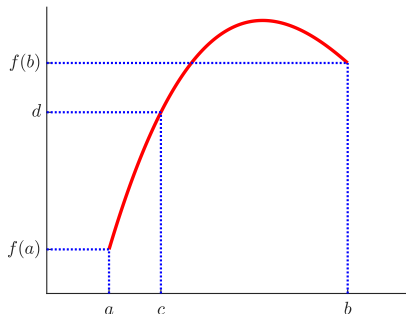
We introduce the open interval spanned by α and β

$$\langle \alpha, \beta \rangle := \left(\min\{\alpha, \beta\}, \max\{\alpha, \beta\} \right).$$

We recall the so-called Intermediate Value Theorem, which states that a continuous function takes any value in between two function values.

Theorem (Intermediate Value Theorem):

Let $f \in C[a, b]$, $f(a) \neq f(b)$, and let $d \in \langle f(a), f(b) \rangle$. Then there exists $c \in (a, b)$ such that $f(c) = d$.



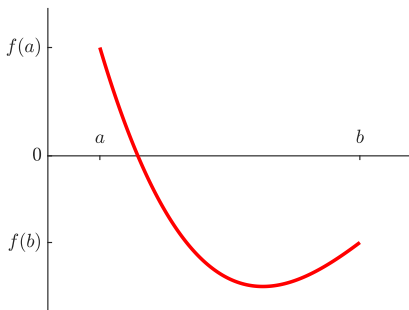
First we study the **bisection method** to solve the scalar nonlinear equation

$$f(x) = 0.$$

We suppose that $f: [a, b] \rightarrow \mathbb{R}$ is a continuous function with opposite sign at the end of the interval, i.e.,

$$f(a)f(b) < 0.$$

Then the Intermediate Value Theorem yields that f has at least one root inside the interval $[a, b]$.



We define a sequence of intervals: Let

$$[a_0, b_0] = [a, b],$$

and let p_0 be the midpoint of the interval, i.e.,

$$p_0 = \frac{a_0 + b_0}{2}.$$

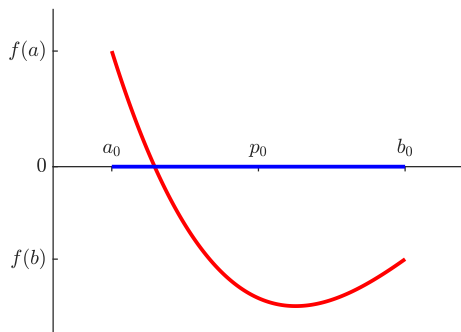
Then either $f(p_0) = 0$, or one of the intervals $[a_0, p_0]$ or $[p_0, b_0]$ has the opposite sign property. If f changes sign on the interval $[a_0, p_0]$, then we define

$$[a_1, b_1] = [a_0, p_0],$$

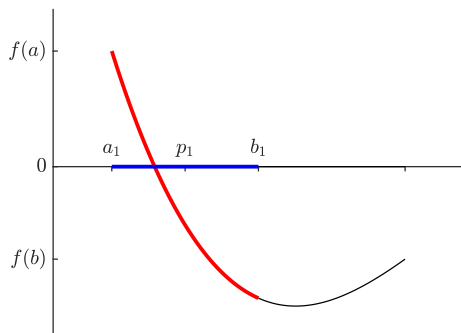
otherwise let

$$[a_1, b_1] = [p_0, b_0].$$

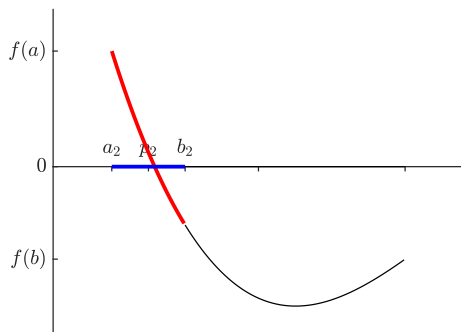
Continuing this procedure, either after finitely many steps, p_k is a root of the function f , or we define an infinite sequence of nested closed bounded intervals $[a_k, b_k]$, so that a root of f is contained in each of the intervals.



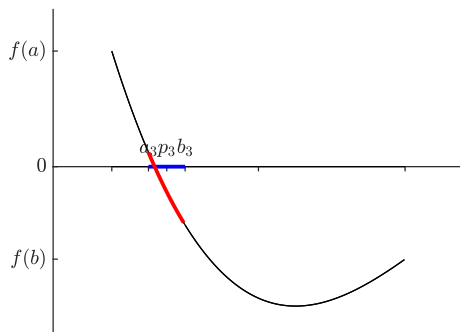
$$n = 0$$



$$n = 1$$



$$n = 2$$



$$n = 3$$

We have that the length of the k th interval

$$b_k - a_k = \frac{b - a}{2^k} \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Then the Cantors's nested intervals theorem shows that there exists $p \in [a, b]$ such that

$$a_k \rightarrow p \quad \text{and} \quad b_k \rightarrow p \quad \text{as } k \rightarrow \infty,$$

and p is the only common point of the intervals. So, in particular, the sequence of midpoints

$$p_k \rightarrow p.$$

Suppose, e.g., that

$$f(a) > 0 \quad \text{and} \quad f(b) < 0.$$

Then for all k

$$f(a_k) > 0 \quad \text{and} \quad f(b_k) < 0.$$

Since

$$a_k \rightarrow p \quad \text{and} \quad b_k \rightarrow p,$$

the continuity of f implies

$$f(p) \geq 0 \quad \text{and} \quad f(p) \leq 0,$$

hence $f(p) = 0$. Since $a_k \leq p \leq b_k$ is satisfied for all k , and

$$\begin{array}{ccccccc} & | & & | & & | & \\ & | & & | & & | & \\ a_k & & p_k & & p & & b_k \end{array}$$

we get

$$|p_k - p| \leq \frac{b_k - a_k}{2} = \frac{b - a}{2^{k+1}}.$$

Theorem:

Let $f \in C[a, b]$ and $f(a)f(b) < 0$. Then the bisection sequence p_k converges to a root p of the function f , and

$$|p_k - p| \leq \frac{b - a}{2^{k+1}}. \quad (4)$$

It follows from the estimate (4) that if we predefine a tolerance (error bound) $\varepsilon > 0$, then

$$|p_k - p| \leq \frac{b - a}{2^{k+1}} < \varepsilon$$

holds if k satisfies

$$k \geq \log_2 \frac{b - a}{\varepsilon} - 1.$$

Example

Consider the function

$$f(x) = e^x - 2 \cos x.$$

Then we have

$$f(0) = -1 \quad \text{and} \quad f(1) > 0,$$

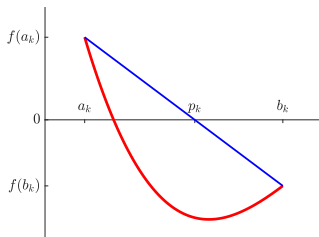
therefore f has a root in the interval $[0, 1]$, and the bisection method is applicable. (It is easy to check that f is strictly monotone increasing on $[0, 1]$, so it has a unique root inside the interval. The next table contains the result of the bisection method using tolerance value $\varepsilon = 10^{-5}$. We have that $k \geq \log_2 10^5 - 1 \approx 15.61$ steps are needed to obtain this accuracy.

k	a_k	b_k	p_k	$f(p_k)$
0	0.00000000	1.00000000	0.50000000	-1.0644e-01
1	0.50000000	1.00000000	0.75000000	6.5362e-01
2	0.50000000	0.75000000	0.62500000	2.4632e-01
3	0.50000000	0.62500000	0.56250000	6.3206e-02
4	0.50000000	0.56250000	0.53125000	-2.3292e-02
5	0.53125000	0.56250000	0.54687500	1.9538e-02
6	0.53125000	0.54687500	0.53906250	-1.9818e-03
7	0.53906250	0.54687500	0.54296875	8.7517e-03
8	0.53906250	0.54296875	0.54101563	3.3784e-03
9	0.53906250	0.54101563	0.54003906	6.9670e-04
10	0.53906250	0.54003906	0.53955078	-6.4294e-04
11	0.53955078	0.54003906	0.53979492	2.6780e-05
12	0.53955078	0.53979492	0.53967285	-3.0810e-04
13	0.53967285	0.53979492	0.53973389	-1.4067e-04
14	0.53973389	0.53979492	0.53976440	-5.6946e-05
15	0.53976440	0.53979492	0.53977966	-1.5083e-05
16	0.53977966	0.53979492	0.53978729	5.8483e-06

2.4 Method of False Position

method of false position (Regula Falsi):

Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous and $f(a)f(b) < 0$. First define $[a_0, b_0] = [a, b]$. At the k th step, let p_k be the intersection of the secant line of f corresponding to the points a_k and b_k and the x -axis.



Little calculation gives that

$$p_k = a_k - f(a_k) \frac{a_k - b_k}{f(a_k) - f(b_k)}. \quad (5)$$

The next interval $[a_{k+1}, b_{k+1}]$ will be either $[a_k, p_k]$ or $[p_k, b_k]$ where the function has sign change.

Algorithm: method of false position

INPUT: f - is a function,
 $[a, b]$ - is an interval, where $f(a)f(b) < 0$
 TOL - is the tolerance,
 $MAXIT$ - is the maximal iteration step,
 OUTPUT: p - is the approximating root.

```

 $i \leftarrow 1$       (step counter)
 $q \leftarrow a$ 
while  $i < MAXIT$  do
     $p \leftarrow a - f(a)(a - b)/(f(a) - f(b))$ 
    if  $|p - q| < TOL$  do
        output( $p$ )
        stop
    end do
  
```

Algorithm: method of false position (cont.)

```
    if  $f(p)f(b) < 0$  do
         $a \leftarrow p$ 
    else if  $f(a)f(p) < 0$  do
         $b \leftarrow p$ 
    else
        output( $p$ )
        stop
    end do
     $i \leftarrow i + 1$ 
     $q \leftarrow p$ 
end do
output(Maximal iteration step is exceeded.)
```

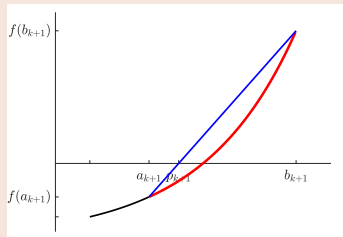
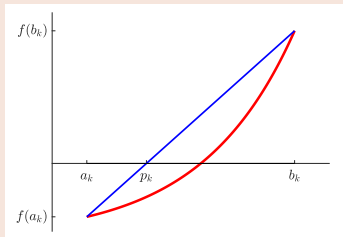
Theorem:

Suppose the continuous function $f \in C[a, b]$ is convex or concave on $[a, b]$ and $f(a)f(b) < 0$. Then the method of false position converges to the unique root p of f .

Proof

Suppose, e.g., that f is convex and $f(a) < 0$, $f(b) > 0$. Then

$$a_{k+1} = p_k \quad \text{and} \quad b_{k+1} = b \quad \text{for all } k.$$



Since the sequence p_k is monotone increasing and $p_k \leq b$, it converges to a limit p , where $a < p < b$.

Proof cont.

Since $f(p_k) < 0$, we get $f(p) \leq 0$. Taking the limit as $k \rightarrow \infty$ in

$$p_k = a_k - f(a_k) \frac{a_k - b_k}{f(a_k) - f(b_k)}$$

we obtain

$$p = p - f(p) \frac{p - b}{f(p) - f(b)},$$

which implies that $f(p) = 0$.

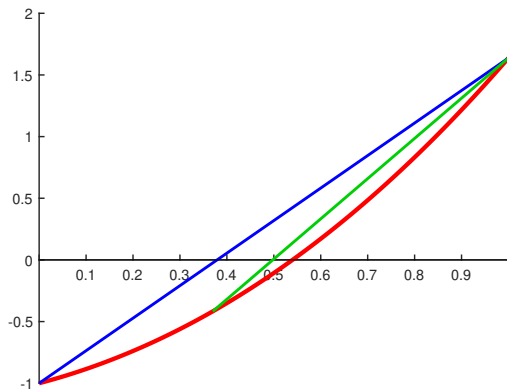
The other cases can be argued similarly.

Example

Solve $e^x - 2 \cos x = 0$ using the method of false position. As before, we use the interval $[0, 1]$ and $TOL = 10^{-5}$. We can observe that for this equation the method of false position converges much faster than the bisection method.

Method of false position, $f(x) = e^x - 2 \cos x$, $[0, 1]$, $TOL = 10^{-5}$

k	a_k	b_k	p_k	$f(p_k)$
0	0.00000000	1.00000000	0.37912145	-3.9698e-01
1	0.37912145	1.00000000	0.50026042	-1.0576e-01
2	0.50026042	1.00000000	0.53057677	-2.5118e-02
3	0.53057677	1.00000000	0.53766789	-5.8011e-03
4	0.53766789	1.00000000	0.53929982	-1.3311e-03
5	0.53929982	1.00000000	0.53967399	-3.0499e-04
6	0.53967399	1.00000000	0.53975970	-6.9856e-05
7	0.53975970	1.00000000	0.53977933	-1.5999e-05
8	0.53977933	1.00000000	0.53978383	-3.6640e-06



$$f(x) = e^x - 2 \cos x$$

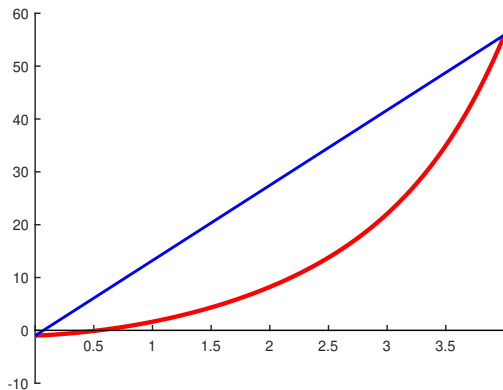
Example

We apply the method of false position for the same equation, but using initial interval $[0, 4]$.

Method of false position, $f(x) = e^x - 2 \cos x$, $[0, 4]$, $TOL = 10^{-5}$

k	a_k	b_k	p_k	$f(p_k)$
0	0.00000000	4.00000000	0.07029205	-9.2224e-01
1	0.07029205	4.00000000	0.13406612	-8.3858e-01
2	0.13406612	4.00000000	0.19119837	-7.5285e-01
3	0.19119837	4.00000000	0.24180834	-6.6826e-01
4	0.24180834	4.00000000	0.28620106	-5.8729e-01
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
47	0.53966897	4.00000000	0.53968870	-2.6464e-04
48	0.53968870	4.00000000	0.53970508	-2.1970e-04
49	0.53970508	4.00000000	0.53971868	-1.8240e-04
50	0.53971868	4.00000000	0.53972996	-1.5143e-04
51	0.53972996	4.00000000	0.53973934	-1.2572e-04

Note, the bisection method on $[0, 4]$ has this accuracy in 18 steps.



$$f(x) = e^x - 2 \cos x$$

2.5 Newton's Method

We recall Taylor's Theorem from calculus.

Theorem (Taylor's Theorem):

Let $f \in C^{n+1}[a, b]$, and let $x_0 \in (a, b)$. Then for every $x \in (a, b)$, $x \neq x_0$ there exists

$$\xi = \xi(x) \in \langle x, x_0 \rangle = \left(\min\{x, x_0\}, \max\{x, x_0\} \right)$$

such that

$$f(x) = T_n(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}, \quad x \in [a, b],$$

where

$$T_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n.$$

Here our goal is to find the solution of the scalar equation

$$f(x) = 0.$$

Fix p_0 , and consider the first-order Taylor approximation of f at p_0 :

$$f(p_0) + f'(p_0)(x - p_0) = 0$$

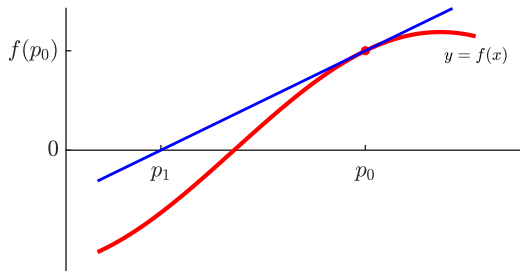
Its solution is

$$x = p_0 - \frac{f(p_0)}{f'(p_0)},$$

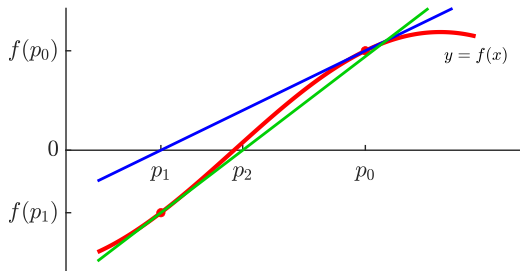
assuming $f'(p_0) \neq 0$. We define the recursive sequence

$$p_{k+1} = p_k - \frac{f(p_k)}{f'(p_k)}. \quad (6)$$

The iterative method (6) is called **Newton–Raphson method** or shortly **Newton's method**.



Newton's method: step 1



Newton's method: step 2

Example

Solve $e^x - 2 \cos x = 0$ using Newton's method. We observe that the sequence converges very fast to the root of the function.

Newton's method, $f(x) = e^x - 2 \cos x$, $p_0 = 0.1$, $TOL = 10^{-5}$

k	p_k	$f(p_k)$
0	0.1000000000	-8.8484e-01
1	0.7781206411	7.5291e-01
2	0.5678850726	7.8450e-02
3	0.5402639121	1.3139e-03
4	0.5397853041	3.9302e-07
5	0.5397851608	3.5207e-14

The Newton's method is a one-step iteration with the function

$$g(x) := x - \frac{f(x)}{f'(x)}. \quad (7)$$

Computing the derivative of g we get

$$g'(x) = 1 - \frac{(f'(x))^2 - f(x)f''(x)}{(f'(x))^2} = \frac{f(x)f''(x)}{(f'(x))^2}. \quad (8)$$

Let p be a root f satisfying $f'(p) \neq 0$. Then

$$g'(p) = 0,$$

so the local version of the fixed point theorem yields immediately the following result.

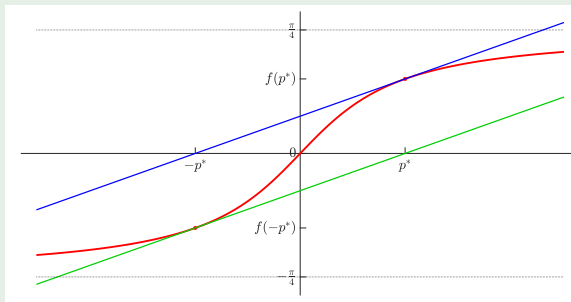
Theorem:

Let $f \in C^2[a, b]$, and let $p \in (a, b)$ be such that $f(p) = 0$ and $f'(p) \neq 0$. Then the Newton's method converges locally to p .

Example

Consider $f(x) = 0.5 \arctan x$. It's only root is $p = 0$. We have that $f'(0) = 0.5$, so the Newton's method converges locally to $p = 0$, i.e., if p_0 is close enough to 0, then the Newton-iteration converges to 0.

It is possible to check that there exists $p^* \approx 1.3918$ such that for $p_0 = p^*$ the sequence is $p^*, -p^*, p^*, -p^*, \dots$, i.e., it is periodic. Moreover, for $|p_0| < p^*$ the sequence $p_n \rightarrow 0$, and for $|p_0| > p^*$ the sequence $|p_n| \rightarrow \infty$.



Next we present a few terms of the Newton-iteration for $p_0 = 1.4$.

Newton's method, $f(x) = 0.5 \arctan x$, $p_0 = 1.4$

k	p_k	$f(p_k)$
0	1.4000000e+00	0.4752734
1	-1.4136186e+00	-0.4775591
2	1.4501293e+00	0.4835443
3	-1.5506260e+00	-0.4990071
4	1.8470541e+00	0.5372889
5	-2.8935624e+00	-0.6190257
6	8.7103258e+00	0.7282453
7	-1.0324977e+02	-0.7805557
8	1.6540564e+04	0.7853679
9	-4.2972148e+08	-0.7853982
10	2.9006412e+17	0.7853982
11	-1.3216239e+35	-0.7853982
12	2.7436939e+70	0.7853982
13	-1.1824729e+141	-0.7853982
14	2.1963537e+282	0.7853982

The Newton's Method is fast, but it requires the computation of $f'(x)$. It is problematic if

- the formula of f is long, then the formula of f' can be very long,
- we do not have a formula for f , but we can evaluate $f(x)$ with a good precision.

2.6 Secant Method

Let p_0 and p_1 be two different initial values of the sequence. Consider the secant line of f corresponding to the points p_0 and p_1 , i.e., the line which connects the points $(p_0, f(p_0))$ and $(p_1, f(p_1))$. Its equation is

$$y = f(p_1) + \frac{f(p_1) - f(p_0)}{p_1 - p_0}(x - p_1).$$

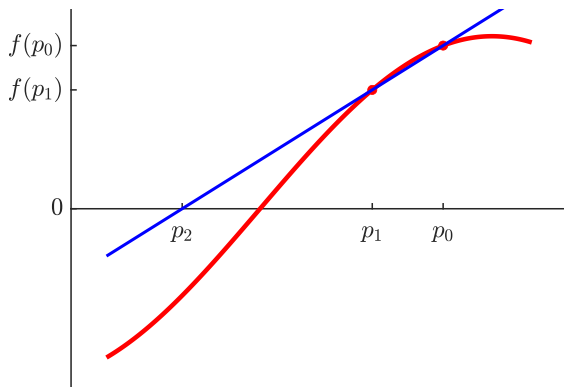
The secant line intersects the x -axis at

$$x = p_1 - \frac{p_1 - p_0}{f(p_1) - f(p_0)}f(p_1).$$

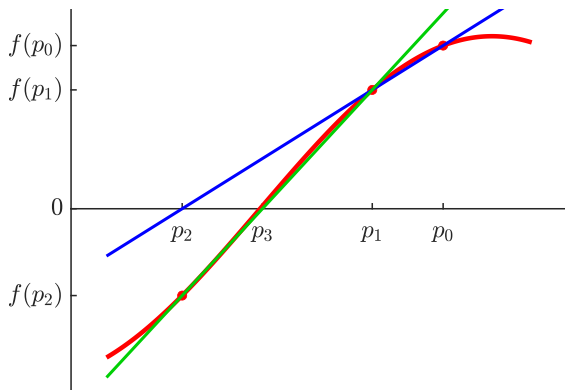
We define the sequence p_k by the recursion

$$p_{k+1} = p_k - \frac{p_k - p_{k-1}}{f(p_k) - f(p_{k-1})}f(p_k). \quad (9)$$

This is a two-step iteration, which defines the **secant method**.



secant method: step 1



secant method: step 2

Example

Solve $e^x - 2 \cos x = 0$ using the secant method. We observe that the secant method converges to its limit slower than the Newton's method.

secant method, $f(x) = e^x - 2 \cos x$, $p_0 = 0$, $p_1 = 1$, $TOL = 10^{-5}$

k	p_k	$f(p_k)$
0	0.0000000000	-1.0000e+00
1	1.0000000000	1.6377e+00
2	0.3791214458	-3.9698e-01
3	0.5002604213	-1.0576e-01
4	0.5442561500	1.2301e-02
5	0.5396724494	-3.0921e-04
6	0.5397848464	-8.6246e-07
7	0.5397851608	6.0793e-11

Theorem:

Let $f \in C^2[a, b]$, and let $p \in (a, b)$ be such that $f(p) = 0$ and $f'(p) \neq 0$. Then the secant method converges locally to p .

2.7 Order of Convergence

Let p_k be a convergent sequence with limit p . We say that the **order of convergence** of the sequence p_k is α if $\alpha \geq 1$ and there exists a constant $c \geq 0$ such that

$$|p_{k+1} - p| \leq c|p_k - p|^\alpha \quad \text{for all } k \geq 0, \quad (10)$$

and if $\alpha = 1$, then we also assume that $c < 1$. More precise to say that the order of convergence is **at least** α . An equivalent form of (10) is

$$\frac{|p_{k+1} - p|}{|p_k - p|^\alpha} \leq c, \quad k \geq 0.$$

If we want to emphasize that p_k satisfies (10) with some α , but it does not satisfy it with any exponent bigger than α , then we say that the order of convergence is **exactly** α .

If the order of convergence of a sequence is $\alpha = 1$, then we say that the convergence is **linear**, and if $\alpha = 2$, then we say that the convergence is **quadratic**.

Suppose p_k converges to p linearly. Then

$$|p_k - p| \leq c|p_{k-1} - p| \leq c^2|p_{k-2} - p| \leq \cdots \leq c^k|p_0 - p|.$$

For some cases, it is not easy to show a linear convergence of a numerical method using the definition (10). So we extend the previous definition so that, if a sequence satisfies relation

$$|p_k - p| \leq c^k|p_0 - p|.$$

with a constant $0 \leq c < 1$, then we also say that the **convergence is linear**.

Suppose $p_k \rightarrow p$ with order α . If the finite limit

$$\lambda = \lim_{k \rightarrow \infty} \frac{p_{k+1} - p}{(p_k - p)^\alpha} \quad (11)$$

exists, then we call λ as the **asymptotic error constant**.

If the limit (11) exists and it is finite, then p_k is convergent and its order of convergence is α .

If p_k converges linearly and its asymptotic error constant is 0, then we speak about **superlinear** convergence.

Theorem:

Suppose p_k converges to p of order α with the asymptotic error constant $\lambda \neq 0$. Then

- (i) $\lim_{k \rightarrow \infty} \frac{p_{k+1} - p}{(p_k - p)^\beta} = 0$ for all $\beta < \alpha$, and
- (ii) $\lim_{k \rightarrow \infty} \frac{|p_{k+1} - p|}{|p_k - p|^\beta} = \infty$ for all $\beta > \alpha$.

Proof

The statements follow from relation

$$\frac{|p_{k+1} - p|}{|p_k - p|^\beta} = \frac{|p_{k+1} - p|}{|p_k - p|^\alpha} \frac{1}{|p_k - p|^{\beta-\alpha}}.$$

Therefore, if a sequence p_k converges to p of order α , and the asymptotic error constant $\lambda \neq 0$, then the order of convergence is exactly α .

Example

Consider again the example studied using the Newton's iteration. We have listed in the last three columns the numerical values of the formula $|p_{k+1} - p|/|p_k - p|^\alpha$ for $\alpha = 1, 2$ and 3 using the value $p = 0.5397851608092811$.

Order of convergence of the Newton iteration, $f(x) = e^x - 2 \cos x$

k	p_k	$f(p_k)$	$ p_k - p / p_{k-1} - p ^\alpha$		
			$\alpha = 1$	$\alpha = 2$	$\alpha = 3$
0	0.0000000000	-1.0000e+00			
1	1.0000000000	1.6377e+00	8.5259e-01	1.5795e+00	2.9262e+00
2	0.6279041258	2.5516e-01	1.9147e-01	4.1605e-01	9.0404e-01
3	0.5442066314	1.2164e-02	5.0176e-02	5.6941e-01	6.4619e+00
4	0.5397973257	3.3375e-05	2.7513e-03	6.2226e-01	1.4074e+02
5	0.5397851609	2.5388e-10	7.6071e-06	6.2533e-01	5.1404e+04

Theorem:

Suppose a sequence p_k satisfies inequality (10) with some $c \geq 0$ and $\alpha > 1$. Then p_k converges locally to p , and for every k

$$|p_k - p| \leq c^{\frac{\alpha^k - 1}{\alpha - 1}} |p_0 - p|^{\alpha^k}. \quad (12)$$

Proof

$$\begin{aligned} |p_k - p| &\leq c |p_{k-1} - p|^\alpha \leq c \left(c |p_{k-2} - p|^\alpha \right)^\alpha = c^{1+\alpha} |p_{k-2} - p|^{\alpha^2} \\ &\leq c^{1+\alpha+\alpha^2} |p_{k-3} - p|^{\alpha^3} \leq \dots \leq c^{1+\alpha+\alpha^2+\dots+\alpha^{k-1}} |p_0 - p|^{\alpha^k}, \end{aligned}$$

which yields (12). Then it implies

$$|p_k - p| \leq c^{\frac{1}{1-\alpha}} \left(c^{\frac{1}{\alpha-1}} |p_0 - p| \right)^{\alpha^k}.$$

Hence if p_0 is such that $c^{\frac{1}{\alpha-1}} |p_0 - p| < 1$, then $p_k \rightarrow p$, i.e., p_k converges locally to p .

Example

Suppose $p_k \rightarrow p$ and $q_k \rightarrow q$ linearly and quadratically, respectively, with $c = 1/2$. Moreover, we suppose $|p_0 - p| < 1$ and $|q_0 - q| < 1$. Then the definition of the order of convergence yields $|p_k - p| \leq (1/2)^k$ and $|q_k - q| \leq (1/2)^{2^k - 1}$. In the next table we listed these error bounds for $k = 1, 2, \dots, 5$. We can see that the error decreases much faster in the quadratic case.

k	$(1/2)^k$	$(1/2)^{2^k - 1}$
1	$5.0000 \cdot 10^{-1}$	$5.0000 \cdot 10^{-1}$
2	$2.5000 \cdot 10^{-1}$	$1.2500 \cdot 10^{-1}$
3	$1.2500 \cdot 10^{-1}$	$7.8125 \cdot 10^{-3}$
4	$6.2500 \cdot 10^{-2}$	$3.0518 \cdot 10^{-5}$
5	$3.1250 \cdot 10^{-2}$	$4.6566 \cdot 10^{-10}$
6	$1.5625 \cdot 10^{-2}$	$1.0842 \cdot 10^{-19}$

Theorem:

Let $g \in C^m[a, b]$, $p \in (a, b)$ and $p = g(p)$. Consider the fixed-point iteration $p_{k+1} = g(p_k)$.

- (i) If $|g'(p)| < 1$, then the fixed-point iteration converges locally and linearly to p .*
- (ii) If $g'(p) = g''(p) = \dots = g^{(m-1)}(p) = 0$, then the fixed-point iteration converges locally to p of order m with the asymptotic error constant $g^{(m)}(p)/m!$.*

Proof

For the proof of statement (ii), we consider the Taylor approximation of g around p with order $(m - 1)$:

$$g(p_k) = g(p) + g'(p)(p_k - p) + \cdots + \frac{g^{(m-1)}(p)}{(m-1)!}(p_k - p)^{m-1} + \frac{g^{(m)}(\xi_k)}{m!}(p_k - p)^m,$$

where $\xi_k \in \langle p_k, p \rangle$. Using that the first $m - 1$ derivatives are equal to 0 at p , $g(p) = p$ and $g(p_k) = p_{k+1}$, we get

$$|p_{k+1} - p| = \frac{|g^{(m)}(\xi_k)|}{m!} |p_k - p|^m \leq c |p_k - p|^m. \quad (13)$$

We used that $g \in C^m[a, b]$, i.e., $g^{(m)}$ is continuous, and therefore, it is bounded in a neighborhood of p . The limit $\xi_k \rightarrow p$ follows from relation $|\xi_k - p| \leq |p_k - p|$. Therefore the asymptotic error constant is

$$\lim_{k \rightarrow \infty} \frac{p_{k+1} - p}{(p_k - p)^m} = \lim_{k \rightarrow \infty} \frac{g^{(m)}(\xi_k)}{m!} = \frac{g^{(m)}(p)}{m!}.$$

We say that $p \in (a, b)$ is a root of **multiplicity m** of $f \in C[a, b]$ if there exists a function $q \in C[a, b]$ such that $q(p) \neq 0$ and

$$f(x) = (x - p)^m q(x), \quad x \in (a, b). \quad (14)$$

Theorem:

Let $f \in C^m[a, b]$, $p \in (a, b)$.

- (i) *Let p be a root of multiplicity m of f , and the function q in (14) is m times differentiable. Then*

$$f(p) = f'(p) = f''(p) = \dots = f^{(m-1)}(p) = 0, \quad \text{and } f^{(m)}(p) \neq 0. \quad (15)$$

- (ii) *If (15) holds, then p is a root of multiplicity m of f .*

Proof

(i) Consider

$$f(x) = (x - p)^m q(x).$$

Then

$$f'(x) = m(x - p)^{m-1}q(x) + (x - p)^m q'(x),$$

so $f'(p) = 0$ if $m > 1$. We get

$$\begin{aligned} f''(x) &= m(m-1)(x-p)^{m-2}q(x) + m(x-p)^{m-1}q'(x) \\ &\quad + m(x-p)^{m-1}q'(x) + (x-p)^m q''(x), \end{aligned}$$

so $f''(p) = 0$ if $m > 2$.

Proof

(ii) Using Taylor's Theorem we get

$$\begin{aligned}
 f(x) &= f(p) + f'(p)(x-p) + \frac{f''(p)}{2}(x-p)^2 + \cdots + \frac{f^{(m-1)}(p)}{(m-1)!}(x-p)^{m-1} \\
 &\quad + \frac{f^{(m)}(\xi(x))}{m!}(x-p)^m \\
 &= (x-p)^m q(x),
 \end{aligned}$$

where

$$q(x) = \frac{f^{(m)}(\xi(x))}{m!}.$$

Theorem:

Let $f \in C^2[a, b]$.

- (i) *If $f(p) = 0$ and $f'(p) \neq 0$, then the Newton iteration converges locally to p , and the order of convergence is quadratic.*
- (ii) *If $f(x) = (x - p)^m q(x)$, where $q \in C^2[a, b]$, $q(p) \neq 0$, $m > 1$, then the Newton iteration converges locally to p , and the order of convergence is linear.*

Proof

Statement (i) follows from earlier results, since the Newton iteration is a fixed-point iteration with the function g defined in (7), and $g'(p) = 0$ by relation (8).

Since the function

$$g(x) := \begin{cases} x - \frac{f(x)}{f'(x)}, & x \neq p, \\ p & x = p \end{cases}$$

satisfies

$$g(x) = x - \frac{(x-p)^m q(x)}{m(x-p)^{m-1}q(x) + (x-p)^m q'(x)} = x - \frac{(x-p)q(x)}{mq(x) + (x-p)q'(x)},$$

it is continuously differentiable at p , and little calculation yields

$$g'(p) = 1 - \frac{1}{m}.$$

Therefore part (ii) of the last theorem yields that the order of convergence is linear.

Example

Find the root of

$$f(x) = x^3 + x^2 - 8x - 12$$

by the Newton–Raphson method from the initial value $p_0 = 0$ and using tolerance 10^{-5} . It is easy to see that $x = -2$ is a double root, and $x = 3$ is a simple root of the polynomial. In the next tables we can see the numerical values of the iteration corresponding to $p_0 = 0$, and corresponding to $p_0 = 2$.

Newton iteration, $f(x) = x^3 + x^2 - 8x - 12$

k	p_k	$f(p_k)$	$ p_k - p / p_{k-1} - p ^\alpha$	
			$\alpha = 1$	$\alpha = 2$
0	0.0000000000	-1.2000e+01		
1	-1.5000000000	-1.1250e+00	2.5000e-01	1.2500e-01
2	-1.7647058824	-2.6379e-01	4.7059e-01	9.4118e-01
3	-1.8853313477	-6.4237e-02	4.8734e-01	2.0712e+00
4	-1.9433465411	-1.5866e-02	4.9406e-01	4.3086e+00
5	-1.9718365260	-3.9436e-03	4.9712e-01	8.7747e+00
6	-1.9859582600	-9.8308e-04	4.9858e-01	1.7703e+01
7	-1.9929890302	-2.4542e-04	4.9929e-01	3.5558e+01
8	-1.9964969780	-6.1313e-05	4.9965e-01	7.1267e+01
9	-1.9982491032	-1.5323e-05	4.9982e-01	1.4268e+02
10	-1.9991247050	-3.8300e-06	4.9991e-01	2.8552e+02
11	-1.9995623908	-9.5743e-07	4.9996e-01	5.7119e+02
12	-1.9997812050	-2.3935e-07	4.9998e-01	1.1425e+03
13	-1.9998906049	-5.9835e-08	4.9999e-01	2.2852e+03
14	-1.9999453030	-1.4959e-08	4.9999e-01	4.5705e+03
15	-1.9999726517	-3.7396e-09	5.0000e-01	9.1412e+03
16	-1.9999863259	-9.3491e-10	5.0000e-01	1.8283e+04
17	-1.9999931629	-2.3373e-10	5.0000e-01	3.6565e+04

Newton iteration, $f(x) = x^3 + x^2 - 8x - 12$

k	p_k	$f(p_k)$	$ p_k - p / p_{k-1} - p ^\alpha$	
			$\alpha = 1$	$\alpha = 2$
0	2.0000000000	-1.6000e+01		
1	4.0000000000	3.6000e+01	1.0000e+00	1.0000e+00
2	3.2500000000	6.8906e+00	2.5000e-01	2.5000e-01
3	3.0217391304	5.4821e-01	8.6957e-02	3.4783e-01
4	3.0001866020	4.6654e-03	8.5837e-03	3.9485e-01
5	3.0000000139	3.4816e-07	7.4632e-05	3.9996e-01
6	3.0000000000	1.9400e-15	5.5721e-09	4.0011e-01

Theorem:

If p is a simple root of f , then the secant method converges locally to p of order $\alpha = (1 + \sqrt{5})/2 \approx 1.618$.

Let $f \in C^3[a, b]$, suppose $p \in (a, b)$ is a multiple root of f . More precisely, we assume that $f(x) = (x - p)^m q(x)$ with $m > 1$ and $q \in C^3[a, b]$. We define the function

$$\mu(x) = \begin{cases} \frac{f(x)}{f'(x)}, & \text{if } x \neq p, \\ 0, & \text{if } x = p. \end{cases}$$

We can see that

$$\mu(x) = \frac{(x - p)q(x)}{mq(x) + (x - p)q'(x)},$$

and hence $\mu \in C^2[a, b]$. Moreover, $\mu'(p) = \frac{1}{m}$, and so p is only a simple root of μ . Therefore if we use the Newton iteration for the function μ instead of f , we get a quadratic convergence. Then we get the sequence

$$p_{k+1} = p_k - \frac{\mu(p_k)}{\mu'(p_k)} = p_k - \frac{f(p_k)f'(p_k)}{(f'(p_k))^2 - f(p_k)f''(p_k)}. \quad (16)$$

2.8 Review of Multivariable Calculus

Let $E \subset \mathbb{R}^n$, and consider the function $f: E \rightarrow \mathbb{R}$ of n variables. The partial derivative of the function

$$f = f(\mathbf{x}) = f(x_1, \dots, x_n)$$

with respect to the variable x_i is denoted by $\frac{\partial f}{\partial x_i}$.

If all the partial derivatives of f up to order m exist and are continuous, then we say that f is m times continuously partially differentiable, and we will denote it by $f \in C^m$.

If $f \in C^1$, then f' denotes the **gradient vector** or shortly, the **gradient** of f :

$$f'(\mathbf{x}) := \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right)^T.$$

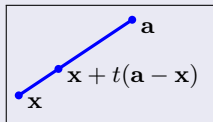
If $f \in C^2$, then $f''(\mathbf{x})$ is the so-called the **Hessian matrix** or shortly the **Hessian**: defined by

$$f''(\mathbf{x}) := \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{pmatrix}$$

Theorem (Taylor's formula):

Let $E \subset \mathbb{R}^n$ be an open set, $f: E \rightarrow \mathbb{R}$, $f \in C^{m+1}$, and let $\mathbf{a} \in E$. Then for every $\mathbf{x} \in E$ there exists a $\xi = \xi(\mathbf{x}) \in E$ such that $\xi = \mathbf{x} + t(\mathbf{a} - \mathbf{x})$ for some $t \in (0, 1)$ (i.e., ξ lies on the line segment connecting $\mathbf{a}$ and $\mathbf{x}$), and

$$\begin{aligned} & f(x_1, \dots, x_n) \\ &= f(a_1, \dots, a_n) + \sum_{i=1}^n \frac{\partial f(a_1, \dots, a_n)}{\partial x_i} (x_i - a_i) \\ &+ \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f(a_1, \dots, a_n)}{\partial x_i \partial x_j} (x_i - a_i)(x_j - a_j) \\ &+ \dots + \frac{1}{m!} \sum_{i_1=1}^n \dots \sum_{i_m=1}^n \frac{\partial^m f(a_1, \dots, a_n)}{\partial x_{i_1} \dots \partial x_{i_m}} (x_{i_1} - a_{i_1}) \dots (x_{i_m} - a_{i_m}) \\ &+ \frac{1}{(m+1)!} \sum_{i_1=1}^n \dots \sum_{i_{m+1}=1}^n \frac{\partial^{m+1} f(\xi_1, \dots, \xi_n)}{\partial x_{i_1} \dots \partial x_{i_{m+1}}} (x_{i_1} - a_{i_1}) \dots (x_{i_{m+1}} - a_{i_{m+1}}). \end{aligned}$$



We can easily check that using the notation of the gradient and the Hessian that for $f \in C^3$ we have the second-order Taylor approximation

$$f(\mathbf{x}) \approx f(\mathbf{a}) + f'(\mathbf{a})^T(\mathbf{x} - \mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^T f''(\mathbf{a})(\mathbf{x} - \mathbf{a}).$$

This justifies the notations f' and f'' for the gradient and the Hessian.

Let $I \subset \mathbb{R}$, $g: I \rightarrow \mathbb{R}^n$. and we denote the component functions of g by g_i , i.e., we use the notation

$$g(t) = (g_1(t), \dots, g_n(t))^T.$$

We say that such g is **differentiable** if all its component functions are differentiable, and its derivative is

$$g': I \rightarrow \mathbb{R}^n, \quad g'(t) := (g'_1(t), \dots, g'_n(t))^T.$$

We say that g is **continuously differentiable** if its each component function is continuously differentiable.

Theorem (chain rule):

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1$ and $g: \mathbb{R} \rightarrow \mathbb{R}^n$ be continuously differentiable. Then the composite function $f \circ g: \mathbb{R} \rightarrow \mathbb{R}$ is also continuously differentiable, and

$$\frac{d}{dt} f(g(t)) = f'(g(t))^T g'(t).$$

We can get the following generalization of the Lagrange's Mean Value Theorem for multivariable functions from the chain rule.

Theorem (Lagrange's Mean Value Theorem):

Let $E \subset \mathbb{R}^n$ be an open and convex set, $f: E \rightarrow \mathbb{R}$ is continuously differentiable with respect to all variables. Then for every $\mathbf{x}, \mathbf{y} \in E$ there exists $\xi \in (0, 1)$ such that

$$f(\mathbf{x}) - f(\mathbf{y}) = f'(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y}))^T (\mathbf{x} - \mathbf{y}).$$

Proof

We define the single variable function $g(t) = f(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))$ for $t \in [0, 1]$. Using the Lagrange's Mean Value Theorem of single variable functions and the chain rule, we get

$$f(\mathbf{x}) - f(\mathbf{y}) = g(1) - g(0) = g'(\xi)(1 - 0) = f'(\mathbf{x} + \xi(\mathbf{y} - \mathbf{x}))^T (\mathbf{x} - \mathbf{y}).$$

Let $E \subset \mathbb{R}^n$ and $\mathbf{f}: E \rightarrow \mathbb{R}^n$. The component functions of $\mathbf{f}$ are denoted by f_i , i.e.,

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_n(\mathbf{x}))^T.$$

We say that $\mathbf{f}$ is **m times continuously partially differentiable** if its every component function is m times continuously differentiable, and it will be denoted by $\mathbf{f} \in C^m$.

The **Jacobian matrix** or shortly, the **Jacobian** of the function $\mathbf{f} \in C^1$ is the $n \times n$ matrix defined by

$$\mathbf{f}'(\mathbf{x}) := \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_1}{\partial x_n}(\mathbf{x}) \\ \vdots & & \vdots \\ \frac{\partial f_n}{\partial x_1}(\mathbf{x}) & \cdots & \frac{\partial f_n}{\partial x_n}(\mathbf{x}) \end{pmatrix}.$$

Let $\mathbf{a} \in \mathbb{R}^n$ be fixed. If we approximate the component functions of $\mathbf{f}$ by its first-order Taylor polynomial around $\mathbf{a}$, then we get

$$\mathbf{f}(\mathbf{x}) = \begin{pmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{pmatrix} \approx \begin{pmatrix} f_1(\mathbf{a}) + f'_1(\mathbf{a})^T(\mathbf{x} - \mathbf{a}) \\ \vdots \\ f_n(\mathbf{a}) + f'_n(\mathbf{a})^T(\mathbf{x} - \mathbf{a}) \end{pmatrix} = \mathbf{f}(\mathbf{a}) + \mathbf{f}'(\mathbf{a})(\mathbf{x} - \mathbf{a}).$$

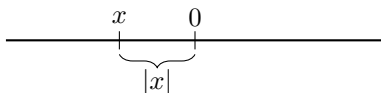
$\mathbf{f}(\mathbf{a}) + \mathbf{f}'(\mathbf{a})(\mathbf{x} - \mathbf{a})$ is called the **linear approximation** of $\mathbf{f}$ around $\mathbf{a}$.

2.9 Vector and Matrix Norms and Convergence

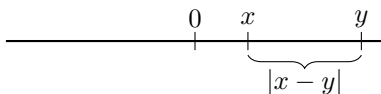
We recall the properties of the absolute value of a real number:

- ① $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$;
- ② $|cx| = |c||x|$, for all $c, x \in \mathbb{R}$;
- ③ $|x + y| \leq |x| + |y|$, for all $x, y \in \mathbb{R}$. (triangle inequality)

The absolute value of x gives back the distance of x from the origin:



The distance between x and y is given by $|x - y|$



The components of the vector $\mathbf{x} \in \mathbb{R}^n$ are denoted by

$$\mathbf{x} = (x_1, x_2, \dots, x_n)^T.$$

Definition

The function $\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R}$ is called **vector norm** if

- ① $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$,
- ② $\|c\mathbf{x}\| = |c|\|\mathbf{x}\|$ for all $c \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$,
- ③ (triangle inequality:) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Theorem:

For any vector norm $\|\cdot\|$ it follows that

- (i) $\left| \|\mathbf{x}\| - \|\mathbf{y}\| \right| \leq \|\mathbf{x} - \mathbf{y}\|,$
- (ii) $\|\cdot\|$ is a continuous function on $\mathbb{R}^n$.

Proof

The triangle inequality yields

$$\|\mathbf{x}\| = \|\mathbf{x} - \mathbf{y} + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\|.$$

Hence we get

$$\|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|.$$

Similarly, $\|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|$ satisfies too, so part (i) follows.

The continuity of $\|\cdot\|$ follows from part (i).

Let $p \geq 1$, and define the so-called **p -norm**:

$$\|\mathbf{x}\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Clearly, $\|\mathbf{x}\|_p \geq 0$, and $\|\mathbf{x}\|_p = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

We have

$$\|c\mathbf{x}\|_p = \left(\sum_{i=1}^n |cx_i|^p \right)^{1/p} = \left(|c|^p \sum_{i=1}^n |x_i|^p \right)^{1/p} = |c| \|\mathbf{x}\|_p.$$

It can be shown that $\|\cdot\|_p$ satisfies the triangle inequality too, i.e., $\|\cdot\|_p$ is a norm for all $p \geq 1$.

The norm corresponding to $p = 2$, i.e.,

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

is called the **Euclidean norm** or **2-norm**.

Another special case is the **1-norm**:

$$\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|.$$

We will also use the following vector norm, the so-called **infinity norm** or **maximum norm**

$$\|\mathbf{x}\|_\infty := \max_{i=1,\dots,n} |x_i|.$$

We have $\|\mathbf{x}\|_\infty \geq 0$, and $\|\mathbf{x}\|_\infty = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Consider

$$\|c\mathbf{x}\|_\infty := \max_{i=1,\dots,n} |cx_i| = \max_{i=1,\dots,n} |c||x_i| = |c| \max_{i=1,\dots,n} |x_i| = |c|\|\mathbf{x}\|_\infty.$$

For a fixed i we have

$$|x_i + y_i| \leq |x_i| + |y_i| \leq \max_{i=1,\dots,n} |x_i| + \max_{i=1,\dots,n} |y_i| = \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty.$$

Since the right-hand side is independent of i , we get

$$\|\mathbf{x} + \mathbf{y}\|_\infty = \max_{i=1,\dots,n} |x_i + y_i| \leq \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty,$$

hence $\|\cdot\|_\infty$ is a norm.

Theorem (Cauchy–Bunyakovsky–Schwarz inequality):

For every $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$ it follows

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i^2,$$

where equality holds if and only if there exists $\lambda \in \mathbb{R}$ such that $y_i = \lambda x_i$ for every $i = 1, 2, \dots, n$.

Proof

Consider the second-order polynomial

$$p(t) := t^2 \sum_{i=1}^n x_i^2 - 2t \sum_{i=1}^n x_i y_i + \sum_{i=1}^n y_i^2.$$

Then

$$p(t) = \sum_{i=1}^n (tx_i - y_i)^2 \geq 0$$

holds for all t , so p may not have two distinct real roots, i.e.,

$$4 \left(\sum_{i=1}^n x_i y_i \right)^2 - 4 \sum_{i=1}^n x_i^2 \sum_{i=1}^n y_i^2 \leq 0.$$

This yields the CBS inequality. p has one real root if and only if its discriminant is 0. $p(t) = 0$ holds for some $t = \lambda$ if and only if $y_i = \lambda x_i$ for all $i = 1, 2, \dots, n$.

Taking a square root for both sides of the Cauchy–Bunyakovsky–Schwarz inequality and using vector notation we get:

Corollary

For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ it follows

$$|\mathbf{x}^T \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2,$$

where the equality is satisfied if and only if there exists $\lambda \in \mathbb{R}$ such that $\mathbf{y} = \lambda \mathbf{x}$.

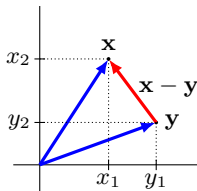
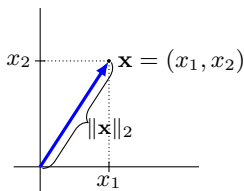
Using the Cauchy–Bunyakovsky–Schwarz inequality we get

$$\begin{aligned}
 \|\mathbf{x} + \mathbf{y}\|_2^2 &= \sum_{i=1}^n (x_i + y_i)^2 \\
 &= \sum_{i=1}^n x_i^2 + 2 \sum_{i=1}^n x_i y_i + \sum_{i=1}^n y_i^2 \\
 &\leq \sum_{i=1}^n x_i^2 + 2 \sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2} + \sum_{i=1}^n y_i^2 \\
 &= \left(\sqrt{\sum_{i=1}^n x_i^2} + \sqrt{\sum_{i=1}^n y_i^2} \right)^2 \\
 &= (\|\mathbf{x}\|_2 + \|\mathbf{y}\|_2)^2,
 \end{aligned}$$

which shows that the Euclidean norm satisfies the triangle inequality.

For two-dimensional vectors $\mathbf{x} = (x_1, x_2)$ we identify the point (x_1, x_2) in a coordinate system with a position vector (a two-dimensional vector with initial point at the origin and terminal point at (x_1, x_2)). Then the length of this vector is

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2}.$$



The distance between the two points $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ is

$$\|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

The expression $\|\mathbf{x}\|$ is called the **length** of the vector, which is the distance between $\mathbf{x}$ and $\mathbf{0}$.

The **distance** between the vectors $\mathbf{x}$ and $\mathbf{y}$ is defined as the real number

$$\|\mathbf{x} - \mathbf{y}\|.$$

Let $\mathbf{p}^{(k)}$ be a sequence of n -dimensional vectors, and let $\|\cdot\|$ be a vector norm on $\mathbb{R}^n$. We say that the sequence $\mathbf{p}^{(k)}$ **converges** to $\mathbf{p}$ if

$$\lim_{k \rightarrow \infty} \|\mathbf{p}^{(k)} - \mathbf{p}\| = 0.$$

Theorem:

Let $|\cdot|$ and $\|\cdot\|$ be two vector norms, and $\mathbf{p}^{(k)}$ be a sequence in $\mathbb{R}^n$. Then $\lim_{k \rightarrow \infty} |\mathbf{p}^{(k)} - \mathbf{p}| = 0$ if and only if $\lim_{k \rightarrow \infty} \|\mathbf{p}^{(k)} - \mathbf{p}\| = 0$.

Proof

It is enough to show that $\|\mathbf{p}^{(k)} - \mathbf{p}\| \rightarrow 0$ if and only if $\|\mathbf{p}^{(k)} - \mathbf{p}\|_1 \rightarrow 0$, where $\|\cdot\|$ is a fixed norm on $\mathbb{R}^n$. It holds if we show that there exist constants m and M such that

$$m\|\mathbf{p}^{(k)} - \mathbf{p}\|_1 \leq \|\mathbf{p}^{(k)} - \mathbf{p}\| \leq M\|\mathbf{p}^{(k)} - \mathbf{p}\|_1. \quad (17)$$

Let $E := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_1 = 1\}$. Then E is a bounded and closed subset of $\mathbb{R}^n$, therefore the continuous function $\|\cdot\|$ takes its maximum and minimum on E . Let denote them by M and m , respectively. Let $\mathbf{x} = (\mathbf{p}^{(k)} - \mathbf{p}) / \|\mathbf{p}^{(k)} - \mathbf{p}\|_1$. Then $\mathbf{x} \in E$, and hence $m \leq \|\mathbf{x}\| \leq M$, which yields (17) after multiplication by $\|\mathbf{p}^{(k)} - \mathbf{p}\|_1$.

Theorem:

Let the i th components of the vectors $\mathbf{p}^{(k)}$ and $\mathbf{p}$ denoted by $p_i^{(k)}$ and p_i , respectively. Then the sequence $\mathbf{p}^{(k)}$ converges to $\mathbf{p}$ if and only if $p_i^{(k)} \rightarrow p_i$ for all $i = 1, 2, \dots, n$ as $k \rightarrow \infty$.

Proof

Applying the last theorem, we have $\|\mathbf{p}^{(k)} - \mathbf{p}\| \rightarrow 0$ if and only if

$$\|\mathbf{p}^{(k)} - \mathbf{p}\|_1 = \sum_{i=1}^n |p_i^{(k)} - p_i| \rightarrow 0,$$

which is satisfied exactly when $p_i^{(k)} \rightarrow p_i$ for all $i = 1, 2, \dots, n$.

The set of $n \times n$ -dimensional real matrices is denoted by $\mathbb{R}^{n \times n}$. Let $\|\cdot\|$ be a vector norm on $\mathbb{R}^n$.

Definition

The function $\|\cdot\|: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ defined by the formula

$$\|\mathbf{A}\| := \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}$$

is called the **matrix norm** generated by the vector norm $\|\cdot\|$.

Here the sup (supremum) denotes the least upper bound, i.e., the smallest M with the property that

$$\frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} \leq M, \quad \text{for all } \mathbf{x} \neq \mathbf{0}.$$

It is possible to show that in the definition of the matrix norm sup can be replaced by max, i.e., there exists a vector $\mathbf{x}$ such that

$$\|\mathbf{A}\| = \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}.$$

Theorem:

For every $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ it follows

- (i) $\|\mathbf{A}\| \geq 0$, and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{0}$,
- (ii) $\|c\mathbf{A}\| = |c|\|\mathbf{A}\|$ for all $c \in \mathbb{R}$,
- (iii) (triangle inequality:) $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$,
- (iv) $\|\mathbf{A}\mathbf{x}\| \leq \|\mathbf{A}\|\|\mathbf{x}\|$, for all $\mathbf{x} \in \mathbb{R}^n$ -re,
- (v) $\|\mathbf{AB}\| \leq \|\mathbf{A}\|\|\mathbf{B}\|$,
- (vi) $\|\mathbf{A}\| = \sup\{\|\mathbf{A}\mathbf{y}\| : \|\mathbf{y}\| = 1\}$.

Proof

$$(ii) \quad \|c\mathbf{A}\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|c\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{|c|\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} = |c| \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|} = |c|\|\mathbf{A}\|$$

$$(iii) \quad \|\mathbf{A} + \mathbf{B}\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|(\mathbf{A} + \mathbf{B})\mathbf{x}\|}{\|\mathbf{x}\|} \leq \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\| + \|\mathbf{B}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| + \|\mathbf{B}\|$$

Proof cont.

Part (iv) follows from

$$\frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} \leq \sup_{\mathbf{y} \neq \mathbf{0}} \frac{\|\mathbf{Ay}\|}{\|\mathbf{y}\|} = \|\mathbf{A}\|.$$

Using (iv) we get

$$\frac{\|\mathbf{ABx}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \frac{\|\mathbf{Bx}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \|\mathbf{B}\|,$$

hence we get (v) from

$$\|\mathbf{AB}\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{ABx}\|}{\|\mathbf{x}\|} \leq \|\mathbf{A}\| \|\mathbf{B}\|.$$

Finally, (vi) follows from

$$\frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|} = \left\| \mathbf{A} \frac{\mathbf{x}}{\|\mathbf{x}\|} \right\|.$$

Theorem:

Let $|\cdot|$ and $\|\cdot\|$ be two vector norms and we consider the corresponding matrix norms. Let $\mathbf{A}^{(k)}$ be a sequence in $\mathbb{R}^{n \times n}$. Then $\lim_{k \rightarrow \infty} |\mathbf{A}^{(k)} - \mathbf{A}| = 0$ if and only if $\lim_{k \rightarrow \infty} \|\mathbf{A}^{(k)} - \mathbf{A}\| = 0$.

Proof

We show that there exist nonnegative constants l and L such that

$$l|\mathbf{B}| \leq \|\mathbf{B}\| \leq L|\mathbf{B}|, \quad \mathbf{B} \in \mathbb{R}^{n \times n}.$$

We know that there exist positive constants m and M such that

$$m|\mathbf{x}| \leq \|\mathbf{x}\| \leq M|\mathbf{x}|, \quad \mathbf{x} \in \mathbb{R}^n.$$

Then

$$\frac{m}{M}|\mathbf{B}| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{m|\mathbf{B}\mathbf{x}|}{M|\mathbf{x}|} \leq \|\mathbf{B}\| = \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{B}\mathbf{x}\|}{\|\mathbf{x}\|} \leq \sup_{\mathbf{x} \neq \mathbf{0}} \frac{M|\mathbf{B}\mathbf{x}|}{m|\mathbf{x}|} = \frac{M}{m}|\mathbf{B}|.$$

Theorem:

Let $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n \times n}$. Then the matrix norms generated by the $\|\cdot\|_1$ and $\|\cdot\|_\infty$ vector norms satisfy

$$\|\mathbf{A}\|_1 = \max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}|,$$

and

$$\|\mathbf{A}\|_\infty = \max_{i=1,\dots,n} \sum_{j=1}^n |a_{ij}|.$$

Proof

Using the definition of the $\|\cdot\|_1$ vector norm and the triangle inequality we get

$$\begin{aligned}
 \|\mathbf{Ax}\|_1 &= \sum_{i=1}^n \left| \sum_{j=1}^n a_{ij} x_j \right| \\
 &\leq \sum_{i=1}^n \sum_{j=1}^n |a_{ij} x_j| \\
 &= \sum_{j=1}^n |x_j| \sum_{i=1}^n |a_{ij}| \\
 &\leq \left(\max_{j=1, \dots, n} \sum_{i=1}^n |a_{ij}| \right) \sum_{j=1}^n |x_j| \\
 &= \left(\max_{j=1, \dots, n} \sum_{i=1}^n |a_{ij}| \right) \|\mathbf{x}\|_1,
 \end{aligned}$$

Proof cont.

hence

$$\|\mathbf{A}\|_1 \leq \max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}|.$$

Suppose

$$\max_{j=1,\dots,n} \sum_{i=1}^n |a_{ij}| = \sum_{i=1}^n |a_{ik}|.$$

We get the statement by multiplying $\mathbf{A}$ and $\mathbf{e}^{(k)} = (0, \dots, 0, 1, 0, \dots, 0)^T$, where $e_i^{(k)} = 0$ if $i \neq k$ and $e_k^{(k)} = 1$. Indeed,

$$\mathbf{A}\mathbf{e}^{(k)} = (a_{1k}, a_{2k}, \dots, a_{nk})^T,$$

therefore

$$\|\mathbf{A}\mathbf{e}^{(k)}\|_1 = \sum_{i=1}^n |a_{ik}|.$$

Theorem:

- ① *If the vector sequence $\mathbf{p}^{(k)}$ is convergent, then its limit is unique.*
- ② *If $\mathbf{p}^{(k)} \rightarrow \mathbf{p}$ and $\mathbf{q}^{(k)} \rightarrow \mathbf{q}$, $\alpha, \beta \in \mathbb{R}$, then the sequence $\alpha\mathbf{p}^{(k)} + \beta\mathbf{q}^{(k)}$ is also convergent, and $\alpha\mathbf{p}^{(k)} + \beta\mathbf{q}^{(k)} \rightarrow \alpha\mathbf{p} + \beta\mathbf{q}$.*
- ③ *If $c_k \rightarrow c$ a real sequence and $\mathbf{p}^{(k)} \rightarrow \mathbf{p}$, then $c_k\mathbf{p}^{(k)} \rightarrow c\mathbf{p}$.*
- ④ *If $\mathbf{p}^{(k)} \rightarrow \mathbf{p}$, then $\mathbf{A}\mathbf{p}^{(k)} \rightarrow \mathbf{A}\mathbf{p}$ for all $\mathbf{A} \in \mathbb{R}^{n \times n}$.*
- ⑤ *(Cauchy's criterion for convergence) $\mathbf{p}^{(k)}$ is a convergent sequence if and only if it is a Cauchy sequence, i.e., for every $\varepsilon > 0$ there exists a $k_0 > 0$ such that $\|\mathbf{p}^{(k)} - \mathbf{p}^{(m)}\| < \varepsilon$ for all $k, m > k_0$.*

Theorem (Lagrange's Mean Value Theorem):

Let $\|\cdot\|$ be a fixed vector norm on $\mathbb{R}^n$, and consider the generated matrix norm. Let $E \subset \mathbb{R}^n$ be an open and convex set, $\mathbf{f}: E \rightarrow \mathbb{R}^n$ be continuously partially differentiable, $\mathbf{x}, \mathbf{y} \in E$. Then

$$\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})\| \leq \max_{t \in [0,1]} \|\mathbf{f}'(\mathbf{y} + t(\mathbf{x} - \mathbf{y}))\| \cdot \|\mathbf{x} - \mathbf{y}\|.$$

Proof

We prove the statement only for the Euclidean norm $\|\cdot\| = \|\cdot\|_2$. Clearly, we can assume that $\mathbf{f}(\mathbf{x}) \neq \mathbf{f}(\mathbf{y})$. Let

$$\mathbf{h} := \frac{\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})}{\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})\|_2}.$$

Then $\|\mathbf{h}\|_2 = 1$. Let $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_n(\mathbf{x}))^T$, $\mathbf{h} = (h_1, \dots, h_n)^T$. We define the real function

$$g(t) := \mathbf{h}^T \mathbf{f}(\mathbf{y} + t(\mathbf{x} - \mathbf{y})) = \sum_{i=1}^n h_i f_i(\mathbf{y} + t(\mathbf{x} - \mathbf{y})).$$

Then, using the Lagrange's Mean Value Theorem for single variable functions and the chain rule, we get

Proof cont.

$$\begin{aligned}
\mathbf{h}^T(\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})) &= g(1) - g(0) \\
&= g'(\xi) \\
&= \sum_{i=1}^n h_i f'_i(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y}))^T (\mathbf{x} - \mathbf{y}) \\
&= \mathbf{h}^T \mathbf{f}'(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y})) (\mathbf{x} - \mathbf{y})
\end{aligned}$$

for some $\xi \in (0, 1)$. Therefore the definition of $\mathbf{h}$, the vector form of the Cauchy-Bunyakovsky-Schwarz inequality and $\|\mathbf{h}\|_2 = 1$ yield

$$\begin{aligned}
\|\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})\|_2 &= \mathbf{h}^T (\mathbf{f}(\mathbf{x}) - \mathbf{f}(\mathbf{y})) \\
&= \mathbf{h}^T \mathbf{f}'(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y})) (\mathbf{x} - \mathbf{y}) \\
&\leq \|\mathbf{h}\|_2 \|f'(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y}))(\mathbf{x} - \mathbf{y})\|_2 \\
&\leq \|f'(\mathbf{y} + \xi(\mathbf{x} - \mathbf{y}))\|_2 \|\mathbf{x} - \mathbf{y}\|_2.
\end{aligned}$$

2.10 Fixed-Point Iteration in n dimensions

Example

Consider the system

$$\begin{aligned} 4x_1 - e^{x_1x_2} - 3 &= 0 \\ x_1 - x_2^2 - 3x_2 - 1 &= 0. \end{aligned} \tag{18}$$

It is easy to check that $x_1 = 1$ and $x_2 = 0$ is a solution of the system. We rearrange (18) in the following way. We express

$$\begin{aligned} x_1 &= \frac{1}{4}(e^{x_1x_2} + 3) \\ x_2 &= \frac{1}{3}(x_1 - x_2^2 - 1) \end{aligned} \tag{19}$$

We can denote the system shortly as $\mathbf{x} = \mathbf{g}(\mathbf{x})$, where $\mathbf{x} = (x_1, x_2)^T$ and

$$\mathbf{g}(\mathbf{x}) = \mathbf{g}(x_1, x_2) = \begin{pmatrix} \frac{1}{4}(e^{x_1x_2} + 3) \\ \frac{1}{3}(x_1 - x_2^2 - 1) \end{pmatrix}. \tag{20}$$

Example cont.

We define an iteration to approximate the solutions of (19) as in the single variable case for $k = 0, 1, 2, \dots$ by

$$\begin{aligned} p_1^{(k+1)} &= \frac{1}{4}(e^{p_1^{(k)}} p_2^{(k)} + 3) \\ p_2^{(k+1)} &= \frac{1}{3} \left(p_1^{(k)} - (p_2^{(k)})^2 - 1 \right). \end{aligned} \tag{21}$$

Defining the vector sequence

$$\mathbf{p}^{(k)} = (p_1^{(k)}, p_2^{(k)})^T,$$

iteration (21) can be written shortly as

$$\mathbf{p}^{(k+1)} = \mathbf{g}(\mathbf{p}^{(k)}).$$

Fixed-point iteration

k	$p_1^{(k)}$	$p_2^{(k)}$
0	-2.000000000	-2.000000000
1	14.399537510	-2.333333333
2	0.750000000	2.651697690
3	2.576641266	-2.427166879
4	0.750480717	-1.438165931
5	0.834956989	-0.772613509
6	0.881152644	-0.253991549
7	0.949867689	-0.061119687
8	0.985899367	-0.017955976
9	0.995613247	-0.004807684
10	0.998806211	-0.001469956
11	0.999633219	-0.000398650
12	0.999900394	-0.000122313

Let $E \subset \mathbb{R}^n$, and consider a function $g: E \rightarrow \mathbb{R}^n$. We say that a vector $\mathbf{p} \in E$ is a **fixed point** of the function g if

$$\mathbf{p} = g(\mathbf{p}).$$

A function $g: E \rightarrow \mathbb{R}^n$ is called a **contraction** on the set E using the vector norm $\|\cdot\|$ if there exists a constant $0 \leq c < 1$ such that

$$\|g(\mathbf{x}) - g(\mathbf{y})\| \leq c\|\mathbf{x} - \mathbf{y}\|$$

for all $\mathbf{x}, \mathbf{y} \in E$.

Theorem (fixed-point theorem):

Let $E \subset \mathbb{R}^n$ be a closed set, $g: E \rightarrow E$, and let g be a contraction on E using a vector norm $\|\cdot\|$. Then g has a unique fixed point $\mathbf{p} \in E$, and the fixed-point iteration $\mathbf{p}^{(k+1)} = g(\mathbf{p}^{(k)})$ converges to $\mathbf{p}$ for all $\mathbf{p}^{(0)} \in E$. The order of convergence is (at least) linear.

Proof

First we show that $\mathbf{p}^{(k)}$ is a Cauchy sequence. Let c be the Lipschitz constant of the function $\mathbf{g}$, and let $k > m$.

$$\begin{aligned}
 & \|\mathbf{p}^{(k)} - \mathbf{p}^{(m)}\| \\
 &= \|\mathbf{p}^{(k)} - \mathbf{p}^{(k-1)} + \mathbf{p}^{(k-1)} - \mathbf{p}^{(k-2)} + \dots + \mathbf{p}^{(m+1)} - \mathbf{p}^{(m)}\| \\
 &\leq \|\mathbf{p}^{(k)} - \mathbf{p}^{(k-1)}\| + \|\mathbf{p}^{(k-1)} - \mathbf{p}^{(k-2)}\| + \dots + \|\mathbf{p}^{(m+1)} - \mathbf{p}^{(m)}\| \\
 &= \|\mathbf{g}(\mathbf{p}^{(k-1)}) - \mathbf{g}(\mathbf{p}^{(k-2)})\| + \|\mathbf{g}(\mathbf{p}^{(k-2)}) - \mathbf{g}(\mathbf{p}^{(k-3)})\| \\
 &\quad + \dots + \|\mathbf{g}(\mathbf{p}^{(m)}) - \mathbf{g}(\mathbf{p}^{(m-1)})\| \\
 &\leq c(\|\mathbf{p}^{(k-1)} - \mathbf{p}^{(k-2)}\| + \|\mathbf{p}^{(k-2)} - \mathbf{p}^{(k-3)}\| + \dots + \|\mathbf{p}^{(m)} - \mathbf{p}^{(m-1)}\|) \\
 &\leq (c^{k-1} + c^{k-2} + \dots + c^m) \|\mathbf{p}^{(1)} - \mathbf{p}^{(0)}\| \\
 &= c^m (c^{k-m-1} + c^{k-m-2} + \dots + 1) \|\mathbf{p}^{(1)} - \mathbf{p}^{(0)}\| \\
 &\leq c^m \sum_{i=0}^{\infty} c^i \|\mathbf{p}^{(1)} - \mathbf{p}^{(0)}\|.
 \end{aligned}$$

Proof cont.

Therefore we get

$$\|\mathbf{p}^{(k)} - \mathbf{p}^{(m)}\| \rightarrow 0 \quad \text{as } m \rightarrow \infty,$$

hence $\mathbf{p}^{(k)}$ is a Cauchy sequence. We get that $\mathbf{p}^{(k)}$ converges to a vector $\mathbf{p}$. Using the continuity of $\mathbf{g}$ we get

$$\begin{array}{ccc} \mathbf{p}^{(k+1)} & = & \mathbf{g}(\mathbf{p}^{(k)}) \\ \downarrow & & \downarrow \\ \mathbf{p} & = & \mathbf{g}(\mathbf{p}) \end{array}$$

i.e., $\mathbf{p}$ is a fixed-point of $\mathbf{g}$.

The order of convergence is at least linear, since

$$\|\mathbf{p}^{(k+1)} - \mathbf{p}\| = \|\mathbf{g}(\mathbf{p}^{(k)}) - \mathbf{g}(\mathbf{p})\| \leq c\|\mathbf{p}^{(k)} - \mathbf{p}\|.$$

Proof cont.

Suppose that $\mathbf{p}$ and $\bar{\mathbf{p}}$ are both fixed points of $\mathbf{g}$. Using the contraction property of $\mathbf{g}$ we have

$$\|\mathbf{p} - \bar{\mathbf{p}}\| = \|\mathbf{g}(\mathbf{p}) - \mathbf{g}(\bar{\mathbf{p}})\| \leq c\|\mathbf{p} - \bar{\mathbf{p}}\|,$$

and therefore $\mathbf{p} = \bar{\mathbf{p}}$ follows.

Theorem:

Let $E \subset \mathbb{R}^n$ be an open set, $\mathbf{g}: E \rightarrow \mathbb{R}^n$, $\mathbf{g} \in C^1$, and let $\mathbf{p}$ be a fixed point of $\mathbf{g}$. If

$$\|\mathbf{g}'(\mathbf{p})\| < 1$$

in a matrix norm generated by a vector norm $\|\cdot\|$, then the fixed-point iteration $\mathbf{p}^{(k+1)} = \mathbf{g}(\mathbf{p}^{(k)})$ locally converges to $\mathbf{p}$.

Proof

Since E is an open set, there exists a radius $\bar{\delta} > 0$ such that

$$\{\mathbf{x}: \|\mathbf{x} - \mathbf{p}\| < \bar{\delta}\} \subset E.$$

Fix a c such that

$$\|\mathbf{g}'(\mathbf{p})\| < c < 1.$$

The function $\mathbf{g}'$ is continuous at $\mathbf{p}$, therefore there exists $0 < \delta \leq \bar{\delta}$ such that

$$\|\mathbf{g}'(\mathbf{x})\| \leq c \quad \text{for all } \mathbf{x} \in V := \{\mathbf{x}: \|\mathbf{x} - \mathbf{p}\| \leq \delta\}.$$

The Lagrange's Mean Value Theorem yields

$$\|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{y})\| \leq \max_{t \in (0,1)} \|\mathbf{g}'(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))\| \cdot \|\mathbf{x} - \mathbf{y}\| \leq c\|\mathbf{x} - \mathbf{y}\|,$$

i.e., $\mathbf{g}$ is a contraction.

Proof cont.

Now we show that the function $\mathbf{g}$ maps the set V into itself. Let $\mathbf{x} \in V$. The contraction property of $\mathbf{g}$ implies

$$\|\mathbf{g}(\mathbf{x}) - \mathbf{p}\| = \|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{p})\| \leq c\|\mathbf{x} - \mathbf{p}\| < \delta,$$

hence

$$\mathbf{g}(\mathbf{x}) \in V.$$

If we restrict $\mathbf{g}$ to the set V , then this function satisfies the conditions of the previous theorem, therefore the fixed-point iteration with initial value from V converges to $\mathbf{p}$. □

Example

Compute the Jacobian matrix of the function

$$\mathbf{g}(\mathbf{x}) = \mathbf{g}(x_1, x_2) = \begin{pmatrix} \frac{1}{4}(e^{x_1 x_2} + 3) \\ \frac{1}{3}(x_1 - x_2^2 - 1) \end{pmatrix}$$

defined by (20):

$$\mathbf{g}'(\mathbf{x}) = \begin{pmatrix} \frac{1}{4}x_2 e^{x_1 x_2} & \frac{1}{4}x_1 e^{x_1 x_2} \\ \frac{1}{3} & -\frac{2}{3}x_2 \end{pmatrix}.$$

Its value at the fixed point of $\mathbf{g}$, i.e., at the point $(1, 0)^T$ is

$$\mathbf{g}'(1, 0) = \begin{pmatrix} 0 & \frac{1}{4} \\ \frac{1}{3} & 0 \end{pmatrix}.$$

Its 1-norm is $\|\mathbf{g}'(1, 0)\|_1 = \frac{1}{3} < 1$, hence our theorem yields that the fixed-point iteration converges locally to $(1, 0)^T$.

Theorem:

Let $E \subset \mathbb{R}^n$, $\mathbf{g}: E \rightarrow \mathbb{R}^n$, $\mathbf{g} \in C^2$, $\mathbf{g}(\mathbf{p}) = \mathbf{p}$, and $\mathbf{g}'(\mathbf{p}) = \mathbf{0}$. Then there exist a $\delta > 0$ such that the fixed-point iteration

$$\mathbf{p}^{(k+1)} = \mathbf{g}(\mathbf{p}^{(k)})$$

converges to $\mathbf{p}$ if

$$\|\mathbf{p}^{(0)} - \mathbf{p}\|_{\infty} < \delta.$$

Moreover, there exists a constant c such that for all k it follows

$$\|\mathbf{p}^{(k+1)} - \mathbf{p}\|_{\infty} \leq c \|\mathbf{p}^{(k)} - \mathbf{p}\|_{\infty}^2,$$

i.e., the iteration locally quadratically converges to $\mathbf{p}$.

Proof

By the assumptions, $0 = \|\mathbf{g}'(\mathbf{p})\| < 1$, therefore the fixed-point iteration is locally convergent.

Consider the second-order Taylor approximation of the i th component function of $\mathbf{g}$ around $\mathbf{p} = (p_1, \dots, p_n)^T$:

$$\begin{aligned} g_i(x_1, \dots, x_n) = & g_i(p_1, \dots, p_n) + \sum_{j=1}^n \frac{\partial g_i(p_1, \dots, p_n)}{\partial x_j} (x_j - p_j) \\ & + \frac{1}{2} \sum_{j=1}^n \sum_{l=1}^n \frac{\partial^2 g_i(\xi_1, \dots, \xi_n)}{\partial x_j \partial x_l} (x_j - p_j)(x_l - p_l). \end{aligned}$$

Applying this relation for $(x_1, \dots, x_n)^T = (p_1^{(k)}, \dots, p_n^{(k)})^T$, and using the relations $p_i = g_i(\mathbf{p})$ and $p_i^{(k+1)} = g_i(\mathbf{p}^{(k)})$, we get

$$p_i^{(k+1)} - p_i = \frac{1}{2} \sum_{j=1}^n \sum_{l=1}^n \frac{\partial^2 g_i(\xi_1, \dots, \xi_n)}{\partial x_j \partial x_l} (p_j^{(k)} - p_j)(p_l^{(k)} - p_l).$$

Proof cont.

Let M be such that $\left| \frac{\partial^2 g_i(x_1, \dots, x_n)}{\partial x_j \partial x_l} \right| \leq M$ for all $i, j, l = 1, \dots, n$ in a neighborhood of $\mathbf{p}$ which contains all $\mathbf{p}^{(k)}$. The definition of M implies

$$|p_i^{(k+1)} - p_i| \leq \frac{1}{2} \sum_{j=1}^n \sum_{l=1}^n M |p_j^{(k)} - p_j| |p_l^{(k)} - p_l| \leq \frac{n^2}{2} M \|\mathbf{p}^{(k)} - \mathbf{p}\|_\infty^2.$$

Since this holds for all $i = 1, \dots, n$, we get

$$\|\mathbf{p}^{(k+1)} - \mathbf{p}\|_\infty \leq \frac{n^2}{2} M \|\mathbf{p}^{(k)} - \mathbf{p}\|_\infty^2,$$

i.e., the order of convergence is quadratic.

2.11 Newton's Method in n dimensions

Let $U \subset \mathbb{R}^n$ an open set, $\mathbf{f}: U \rightarrow \mathbb{R}^n$, consider the nonlinear system

$$\mathbf{f}(\mathbf{x}) = \mathbf{0}.$$

Fix a vector $\mathbf{p}^{(k)} \in U$. As in the scalar case, we approximate $\mathbf{f}$ by its linear part

$$\mathbf{f}(\mathbf{p}^{(k)}) + \mathbf{f}'(\mathbf{p}^{(k)})(\mathbf{x} - \mathbf{p}^{(k)}) = \mathbf{0}.$$

Its root is

$$\bar{\mathbf{x}} = \mathbf{p}^{(k)} - (\mathbf{f}'(\mathbf{p}^{(k)}))^{-1} \mathbf{f}(\mathbf{p}^{(k)}).$$

Therefore we define the **Newton's method** by the iteration

$$\mathbf{p}^{(k+1)} = \mathbf{p}^{(k)} - \left(\mathbf{f}'(\mathbf{p}^{(k)}) \right)^{-1} \mathbf{f}(\mathbf{p}^{(k)}). \quad (22)$$

Theorem:

Let $\mathbf{f} \in C^2$, $\mathbf{f}(\mathbf{p}) = \mathbf{0}$ and suppose the matrix $\mathbf{f}'(\mathbf{p})$ is invertible. Then the Newton's iteration (22) locally quadratically converges to $\mathbf{p}$.

Proof

The Newton's method is a fixed-point iteration with the iteration function

$$\mathbf{g}(\mathbf{x}) = \mathbf{x} - (\mathbf{f}'(\mathbf{x}))^{-1}\mathbf{f}(\mathbf{x}).$$

Let $(\mathbf{f}'(\mathbf{x}))^{-1} = (b_{ij}(\mathbf{x}))_{n \times n}$. Then

$$\sum_{j=1}^n b_{ij}(\mathbf{x}) \frac{\partial f_j(\mathbf{x})}{\partial x_l} = \delta_{il} := \begin{cases} 1, & i = l, \\ 0, & i \neq l. \end{cases} \quad (23)$$

Consider the i th component of $\mathbf{g}$:

$$g_i(\mathbf{x}) = x_i - \sum_{j=1}^n b_{ij}(\mathbf{x}) f_j(\mathbf{x}).$$

Proof cont.

Taking its partial derivative with respect to x_l we get

$$\frac{\partial g_i(\mathbf{x})}{\partial x_l} = \delta_{il} - \sum_{j=1}^n \left(\frac{\partial b_{ij}(\mathbf{x})}{\partial x_l} f_j(\mathbf{x}) + b_{ij}(\mathbf{x}) \frac{\partial f_j(\mathbf{x})}{\partial x_l} \right).$$

At the point $\mathbf{x} = \mathbf{p}$ we get, using relations $f_j(\mathbf{p}) = 0$ and (23), that

$$\frac{\partial g_i(\mathbf{p})}{\partial x_l} = \delta_{il} - \sum_{j=1}^n b_{ij}(\mathbf{p}) \frac{\partial f_j(\mathbf{p})}{\partial x_l} = 0.$$

Therefore

$$\mathbf{g}'(\mathbf{p}) = \mathbf{0},$$

and hence the iteration is locally quadratically convergent.

Applying formula

$$\mathbf{p}^{(k+1)} = \mathbf{p}^{(k)} - \left(\mathbf{f}'(\mathbf{p}^{(k)}) \right)^{-1} \mathbf{f}(\mathbf{p}^{(k)})$$

we need to compute the inverse of a matrix. Instead of it, in practice, we do the following: Introduce the notation

$$\mathbf{s}^{(k)} := \mathbf{p}^{(k+1)} - \mathbf{p}^{(k)},$$

and rearrange equation into the form

$$\mathbf{f}'(\mathbf{p}^{(k)})\mathbf{s}^{(k)} = -\mathbf{f}(\mathbf{p}^{(k)}).$$

We solve it for $\mathbf{s}^{(k)}$, and let

$$\mathbf{p}^{(k+1)} = \mathbf{p}^{(k)} + \mathbf{s}^{(k)}.$$

Example

Consider the system (18) of the last example. We apply the Newton's method for this system starting from the initial value $(-1.5, -1.5)^T$. The next table lists the numerical result. We observe quick convergence.

Newton's method		
k	$\mathbf{p}^{(k)}$	$\ \mathbf{p}^{(k)} - \mathbf{p}\ _\infty$
0	$(-1.500000000000, -1.500000000000)^T$	2.500000e+00
1	$(-1.250000000000, -0.52120413480)^T$	2.250000e+00
2	$(0.53188386800, -0.10035922100)^T$	4.681161e-01
3	$(0.98873605300, -0.00042581408)^T$	1.126395e-02
4	$(0.99999868610, -0.00000037764)^T$	1.313900e-06

2.12 Quasi-Newton Methods, Broyden's Method

We introduce **quasi-Newton methods** which are defined by

$$\mathbf{p}^{(k+1)} = \mathbf{p}^{(k)} - \left(\mathbf{A}^{(k)}\right)^{-1} \mathbf{f}(\mathbf{p}^{(k)}). \quad (24)$$

Here the matrix

$$\mathbf{A}^{(k)} \approx \mathbf{f}'(\mathbf{p}^{(k)}).$$

Using different approximations, we get different classes of quasi-Newton methods.

One typical approach is to approximate the Jacobian matrix numerically. Let $\mathbf{e}^{(j)} = (0, \dots, 0, 1, 0, \dots, 0)^T$ be the j th standard unit vector, $h > 0$ a small discretization constant, and define the components of $\mathbf{A}^{(k)}$ by the expressions

$$a_{ij}^{(k)} = \frac{f_i(\mathbf{p}^{(k)} + h\mathbf{e}^{(j)}) - f_i(\mathbf{p}^{(k)})}{h}, \quad i, j = 1, \dots, n. \quad (25)$$

The resulting quasi-Newton method is a straightforward generalization of the secant method for the vector case.

Next we introduce an other popular selection of the matrices $\mathbf{A}^{(k)}$. This method is called **Broyden's method**. This can also be considered as the generalization of the secant method for the vector case. For scalar equations the secant method replaces the nonlinear equation

$$f(x) = 0$$

by a linear equation

$$f(p_k) + a_k(x - p_k) = 0,$$

where

$$a_k = \frac{f(p_k) - f(p_{k-1})}{p_k - p_{k-1}}.$$

We replace k by $k + 1$, and we rewrite the equation, we get that a_{k+1} solves the equation

$$a_{k+1}(p_{k+1} - p_k) = f(p_{k+1}) - f(p_k). \quad (26)$$

We will generalize this formula for the vector case.

Select an initial vector $\mathbf{p}^{(0)}$ and an initial matrix $\mathbf{A}^{(0)}$. For the selection of $\mathbf{A}^{(0)}$ we can use different strategies:

- it is possible to use the exact value $\mathbf{A}^{(0)} = \mathbf{f}'(\mathbf{p}^{(0)})$,
- or using the formula (25) we can compute an approximate derivative matrix at $\mathbf{p}^{(0)}$,
- or just select any invertible matrix $\mathbf{A}^{(0)}$.

Suppose $\mathbf{p}^{(k)}$ and $\mathbf{A}^{(k)}$ are already defined. Then we define $\mathbf{p}^{(k+1)}$ by formula

$$\mathbf{p}^{(k+1)} = \mathbf{p}^{(k)} - \left(\mathbf{A}^{(k)}\right)^{-1} \mathbf{f}(\mathbf{p}^{(k)}).$$

Similarly to equation (26), we require that $\mathbf{A}^{(k+1)}$ satisfies the so-called **secant equation**

$$\mathbf{A}^{(k+1)}(\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}) = \mathbf{f}(\mathbf{p}^{(k+1)}) - \mathbf{f}(\mathbf{p}^{(k)}). \quad (27)$$

We introduce the following notations

$$\mathbf{y}^{(k)} := \mathbf{f}(\mathbf{p}^{(k+1)}) - \mathbf{f}(\mathbf{p}^{(k)}) \quad \text{and} \quad \mathbf{s}^{(k)} := \mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}.$$

Using this notations, equations (24) and (27) are equivalent to

$$\mathbf{A}^{(k)} \mathbf{s}^{(k)} = -\mathbf{f}(\mathbf{p}^{(k)}), \quad (28)$$

and

$$\mathbf{A}^{(k+1)} \mathbf{s}^{(k)} = \mathbf{y}^{(k)}, \quad (29)$$

respectively. First we solve (28) for $\mathbf{s}^{(k)}$ (assuming that $\mathbf{A}^{(k)}$ is invertible), so the problem is reduced to the selection of a matrix $\mathbf{A}^{(k+1)}$ which satisfies equation (29). Unfortunately, this equation does not determine the matrix $\mathbf{A}^{(k+1)}$ uniquely, since this equation is equivalent to n number of scalar equations, but $\mathbf{A}^{(k+1)}$ is determined by n^2 number of components.

Equation (29) requires that the linear operator $\mathbf{A}^{(k+1)}$ is defined on the one dimensional space spanned by the vector $\mathbf{s}^{(k)}$. But in the $n - 1$ directions orthogonal to the vector $\mathbf{s}^{(k)}$ the linear map is undetermined. Since in the $k + 1$ -th step we “do not have new information” about the next linear operator, i.e., the next matrix, we define $\mathbf{A}^{(k+1)}$ so that its effect on this subspace be the same as the matrix $\mathbf{A}^{(k)}$. Therefore, in addition to equation (29), we require

$$\mathbf{A}^{(k+1)}\mathbf{z} = \mathbf{A}^{(k)}\mathbf{z}, \quad \text{for all } \mathbf{z} \perp \mathbf{s}^{(k)}. \quad (30)$$

Equations (29) and (30) together determine the matrix $\mathbf{A}^{(k+1)}$ uniquely. It can be checked easily that the matrix

$$\mathbf{A}^{(k+1)} = \mathbf{A}^{(k)} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)}\mathbf{s}^{(k)})(\mathbf{s}^{(k)})^T}{\|\mathbf{s}^{(k)}\|_2^2} \quad (31)$$

satisfies both (29) and (30).

To prove (29) consider:

$$\begin{aligned}
 \mathbf{A}^{(k+1)} \mathbf{s}^{(k)} &= \left(\mathbf{A}^{(k)} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)})(\mathbf{s}^{(k)})^T}{\|\mathbf{s}^{(k)}\|_2^2} \right) \mathbf{s}^{(k)} \\
 &= \mathbf{A}^{(k)} \mathbf{s}^{(k)} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)}) \left((\mathbf{s}^{(k)})^T \mathbf{s}^{(k)} \right)}{\|\mathbf{s}^{(k)}\|_2^2} \\
 &= \mathbf{y}^{(k)}
 \end{aligned}$$

To prove (30) consider for $\mathbf{z} \perp \mathbf{s}^{(k)}$:

$$\begin{aligned}
 \mathbf{A}^{(k+1)} \mathbf{z} &= \left(\mathbf{A}^{(k)} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)})(\mathbf{s}^{(k)})^T}{\|\mathbf{s}^{(k)}\|_2^2} \right) \mathbf{z} \\
 &= \mathbf{A}^{(k)} \mathbf{z} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)}) \left((\mathbf{s}^{(k)})^T \mathbf{z} \right)}{\|\mathbf{s}^{(k)}\|_2^2} \\
 &= \mathbf{A}^{(k)} \mathbf{z}
 \end{aligned}$$

The recursion (24) requires the computation of $(\mathbf{A}^{(k)})^{-1}$. The next result is an efficient way to compute it.

Theorem (Sherman–Morrison–Woodbury):

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ be invertible. Then the matrix $\mathbf{A} + \mathbf{u}\mathbf{v}^T$ is invertible if and only if $1 + \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u} \neq 0$, and then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^T)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}}{1 + \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u}}$$

holds.

Proof

Let $\gamma \in \mathbb{R}$, and consider

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^T)(\mathbf{A}^{-1} - \gamma \mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}) = \mathbf{I} + \mathbf{u}\mathbf{v}^T \mathbf{A}^{-1} - \gamma \mathbf{u}\mathbf{v}^T \mathbf{A}^{-1} - \gamma \mathbf{u}\mathbf{v}^T \mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}.$$

Since $\mathbf{v}^T \mathbf{A}^{-1} \mathbf{u}$ is a scalar, we can rewrite the above relation as

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^T)(\mathbf{A}^{-1} - \gamma \mathbf{A}^{-1} \mathbf{u} \mathbf{v}^T \mathbf{A}^{-1}) = \mathbf{I} + (1 - \gamma - \gamma \mathbf{v}^T \mathbf{A}^{-1} \mathbf{u}) \mathbf{u}\mathbf{v}^T \mathbf{A}^{-1},$$

which proves the statement.

A little computation and our Theorem give from (31)

$$\begin{aligned}
 (\mathbf{A}^{(k+1)})^{-1} &= \left(\mathbf{A}^{(k)} + \frac{(\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)})(\mathbf{s}^{(k)})^T}{\|\mathbf{s}^{(k)}\|_2^2} \right)^{-1} \\
 &= (\mathbf{A}^{(k)})^{-1} - \frac{(\mathbf{A}^{(k)})^{-1} \left(\frac{\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)}}{\|\mathbf{s}^{(k)}\|_2^2} \right) (\mathbf{s}^{(k)})^T (\mathbf{A}^{(k)})^{-1}}{1 + (\mathbf{s}^{(k)})^T (\mathbf{A}^{(k)})^{-1} \frac{\mathbf{y}^{(k)} - \mathbf{A}^{(k)} \mathbf{s}^{(k)}}{\|\mathbf{s}^{(k)}\|_2^2}} \\
 &= (\mathbf{A}^{(k)})^{-1} - \frac{((\mathbf{A}^{(k)})^{-1} \mathbf{y}^{(k)} - \mathbf{s}^{(k)}) (\mathbf{s}^{(k)})^T (\mathbf{A}^{(k)})^{-1}}{(\mathbf{s}^{(k)})^T (\mathbf{A}^{(k)})^{-1} \mathbf{y}^{(k)}}.
 \end{aligned} \tag{32}$$

Using iteration (32), if $(\mathbf{A}^{(k)})^{-1}$ is known, then only matrix multiplication is needed to compute $(\mathbf{A}^{(k+1)})^{-1}$, so n^2 number of arithmetic operation is enough to generate the next matrix. On the other hand, the matrix inversion needs n^3 number of operation, so here we have an efficient computational method.

It can be shown that the Broyden's method is locally convergent to a root $\mathbf{p}$ of $\mathbf{f}$ if $\mathbf{A}^{(0)}$ is close enough to $\mathbf{f}'(\mathbf{p})$ and the order of convergence is **superlinear**, i.e.,

$$\lim_{k \rightarrow \infty} \frac{\|\mathbf{p}^{(k+1)} - \mathbf{p}\|}{\|\mathbf{p}^{(k)} - \mathbf{p}\|} = 0.$$

Example

Consider again the system (18), and solve it using Broyden's method with $h = 0.001$ and $TOL = 10^{-5}$. The numerical result is shown in the next table. The last column shows that the speed of the convergence is superlinear.

Broyden's method

k	$\mathbf{p}^{(k)}$	$\ \mathbf{p}^{(k)} - \mathbf{p}\ _\infty$	$\frac{\ \mathbf{p}^{(k)} - \mathbf{p}\ _\infty}{\ \mathbf{p}^{(k-1)} - \mathbf{p}\ _\infty}$
0	$(-1.5000000000, -1.5000000000)^T$	2.5000000000	
1	$(-1.2490215360, -0.5215363883)^T$	2.2490215360	0.8996086144
2	$(-0.4968297655, -0.9366983828)^T$	1.4968297660	0.6655471022
3	$(-0.3045368940, -0.3621731989)^T$	1.3045368940	0.8715332389
4	$(0.5414891937, -0.0587408442)^T$	0.4585108063	0.3514740046
5	$(0.9527177435, -0.0515250779)^T$	0.0515250779	0.1123748387
6	$(1.0003263340, 0.0319681269)^T$	0.0319681269	0.6204382061
7	$(1.0000051000, -0.0040567750)^T$	0.0040567750	0.1269006155
8	$(1.0000069210, -0.0000347010)^T$	0.0000347010	0.0085538489
9	$(1.0000001100, 0.0000012682)^T$	0.0000012682	0.0365458110
10	$(1.0000000050, 0.0000000576)^T$	0.0000000576	0.0453865979